REFLOWECONOMY · BUILD PLAN

ReflowEconomy prototype build plan

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade: feed slot 120 mm high so ISO 13857 Table 4 applies (STEM check fix, RFE-DEC-001 D3); RFE-DWG-109 Rev P3; pictures regenerated
0.3	2026-10-02	Amish Chadha	Decisions of 2026-10-02 carried into the pictures and text: feed hood on the chute, self-closing door and gate hinges, padded and taped tray crossing
0.2	2026-10-02	Amish Chadha	Safety stop S7: padded and marked tray crossing, escape width past the open enclosure door (decided by Amish, 2026-10-02)
0.1	2026-09-30	Amish Chadha	First build plan, with pictures by component and step; design made constructable (RFE-DDR-003)

ReflowEconomy prototype build plan

Plan, not yet built. How to fit out the first reference micro-factory, component by component. Building and running it is TRL 4 work. Decisions still to be made are kept in the design decisions register (docs/06-design-decisions.md), not here.

1. What you are building



Figure 1. Every component pulled apart and numbered in build order. The front row is pulled toward you; wall-mounted and overhead parts are lifted.

The prototype is the reference micro-factory itself: a plastics line on a 12.19 x 4.88 m floor, drawn here in two 40 ft shipping containers joined side by side, which is the harder case (in a rented shed of the same size, skip everything about the seam). Material comes in at the intake doors, is weighed and sorted, washed, shredded inside a lined enclosure and dried along the front row, then pressed into sheets or extruded into beams in the hot zone in the back row, under two enclosing hoods that a fan outside the back wall extracts. Figure 1 shows the 24 components in the order you fit them. Eleven are made in a local workshop from steel angle and tube, galvanised sheet, welded mesh, aluminium tread plate and plywood: the floor seam plate, the backboard, hood B, the transfer bridge, the press booth, the sliding sash and its rails, two duct brackets, the fan stand, the shredder enclosure, the drying rack with its fan stand, and the export cage. The three machines (shredder, extruder and sheet press) are built by a fabricator to their published open designs or bought; everything else is bought and fitted. The work is cutting, drilling, welding and bolting steel, riveting sheet, screwing to a container floor and wall, and licensed electrical work. Value-engineering target for the equipment: USD 25,000. Estimated cost of the constructable design: USD 25,300 (USD 300 over the target), plus about USD 9,200 for the container shell.

> **Safety:** This is a workshop with a shredder, surfaces at 190 to 200 °C, melt fumes, water near electricity, a 230 V 40 A supply, lithium cells hidden in collected waste and a high fire load. The container walls are cut only after a structural engineer's beam is in. Only a licensed electrician connects the supply. Nothing is shredded, heated or melted until the safety stops in section 6 are passed, and a qualified safety professional reviews the finished layout before anyone works in it.

2. What changed to make it buildable

The concept layout showed where everything goes; some of its parts overlapped, hung in the air or could not be reached. Each change below keeps what the micro-factory does, and all of them are recorded in decision record RFE-DDR-003, open for Amish's review.

Table 1. Changes from the concept.

Component	The concept had	The buildable design has	Why
Container seam	Side walls removed, nothing along the seam	A seam beam under the roof on the corner posts (engineer's design) and a floor seam plate (Figures 2 and 4)	The roof needs carrying once the walls go; the floor joint is in the aisle
Hood B	A hood hanging in the air, cutting into the extruder	A sheet box bolted to the extruder frame, the barrel through its end (Figure 13)	The frame is the only thing there to carry it
Duct	Through the booth wall and roof, with no inlet in the booth	Under the booth roof, through a sealed collar, with a booth take-off and damper (Figures 18 and 22)	The booth is now actually extracted
Press booth	One opening, the cooling press unreachable	Two openings, one sliding sash, a transfer bridge between the presses (Figures 15, 17 and 20)	The hot mould stays in the booth; only one opening is ever open
Shredder enclosure	A floor the shredder stood in; chute mouth 2.1 m up; door blocked by the tanks	No floor; roof 1.60 m; a feed hood on the chute with its slot 1.70 m up, so a hand cannot reach the cutters; self-closing door on the aisle face (Figures 26 to 28)	It can be built round the machine, fed from the floor and opened
Board and eyewash	On the bare corrugated wall	On a plywood backboard (Figure 7)	Something flat to screw to
Cables	No route	One cable tray, crossing the aisle under the seam beam, padded and taped where it crosses (Figure 9)	One crossing, with the least loss of headroom
Fan box	Floating outside the wall	On a braced stand, with the stack drawn (Figure 25)	It needs a support that does not load the wall
Drying fan	Floating	On a floor stand (Figure 31)	
Export cage	No gate that could open	A double gate on self-closing hinges on the aisle face (Figure 33)	The export doors are only 170 mm away
Bins	Over the painted aisle line	Behind it; the clear aisle widens to 1.57 m	Keeps the aisle clear

3. Making the components

Make and check each component before the assembly step that needs it. Sizes are in millimetres. Positions along the line are measured from the outside face of the intake end wall, and positions across from the outside face of the front wall (the long wall on the intake row side); both walls are taken as 80 thick, so their inside faces are at 80. "Back" is the long wall with the hot zone. Workshop tolerance is 2 mm unless a step says otherwise; drawings do not carry tolerances before TRL 4.

3.1 Building shell and seam beam (container option)

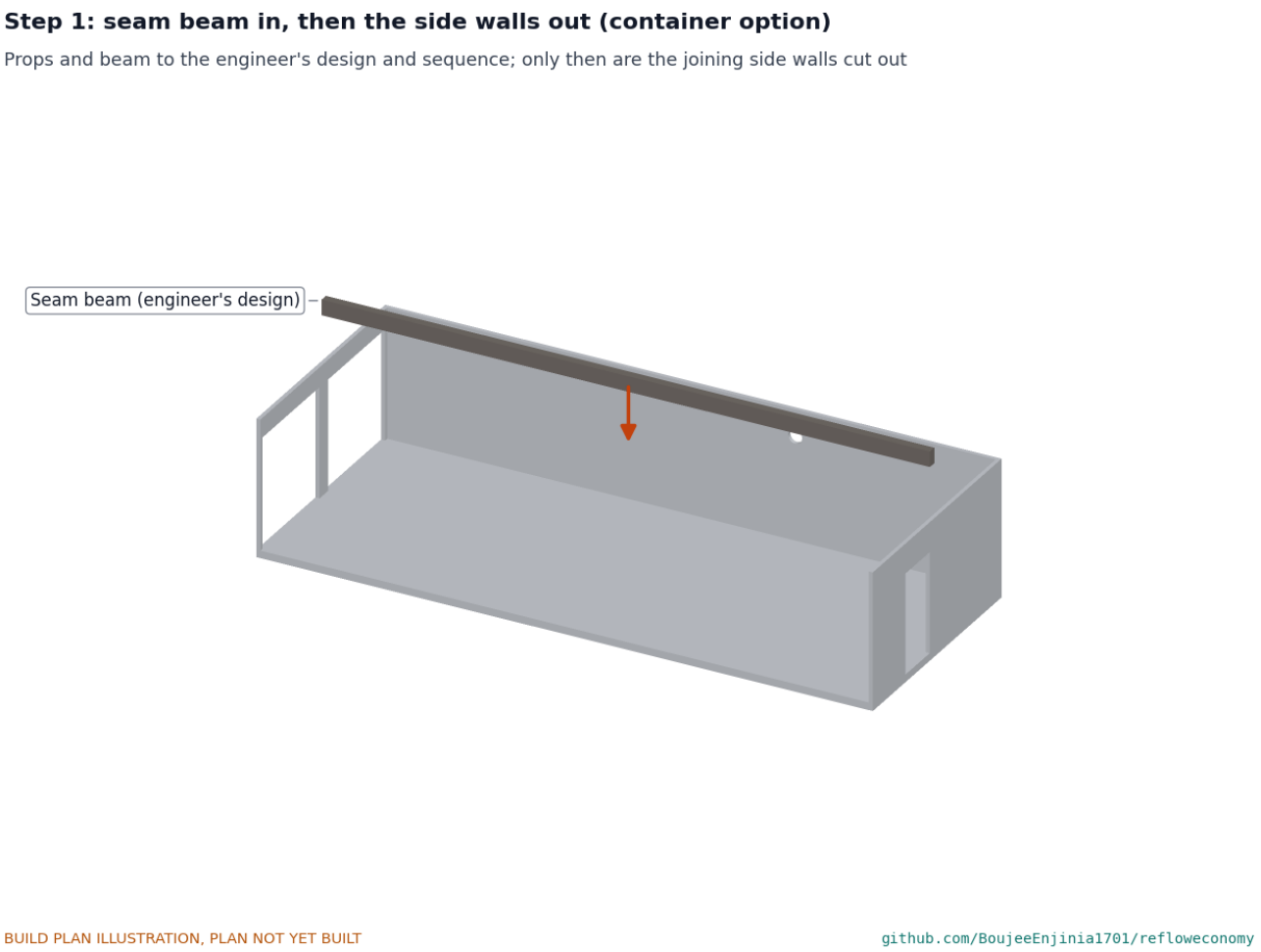


Figure 2. The seam beam goes in under the roof before the joining side walls are cut out.

What it is. Two used 40 ft containers placed side by side on level footings, doors at the intake end, with the two side walls that touch removed so the floor is one room. A personnel exit door is cut in the far end wall of the intake-side container, 1,300 to 2,200 from the front wall and 2,000 high. A 250 mm hole for the fume duct goes through the back wall, centred 8,150 along and 2,050 up.

How it is made. By the container dealer or a fabricator, to a structural engineer's design and sequence: prop the roofs, fit the seam beam under the roof along the seam on the corner posts at both ends, then cut the side walls out, then seal the roof joint with flashing. The design reserves a space 165 wide and 305 deep under the roof along the seam for the beam; its section and fixings are the engineer's. In a rented shed none of this is needed.

Check before moving on. The engineer has signed off the beam and the propping has been removed on their say-so; the roof joint is watertight in rain.

3.2 Floor seam cover plate (make 4)

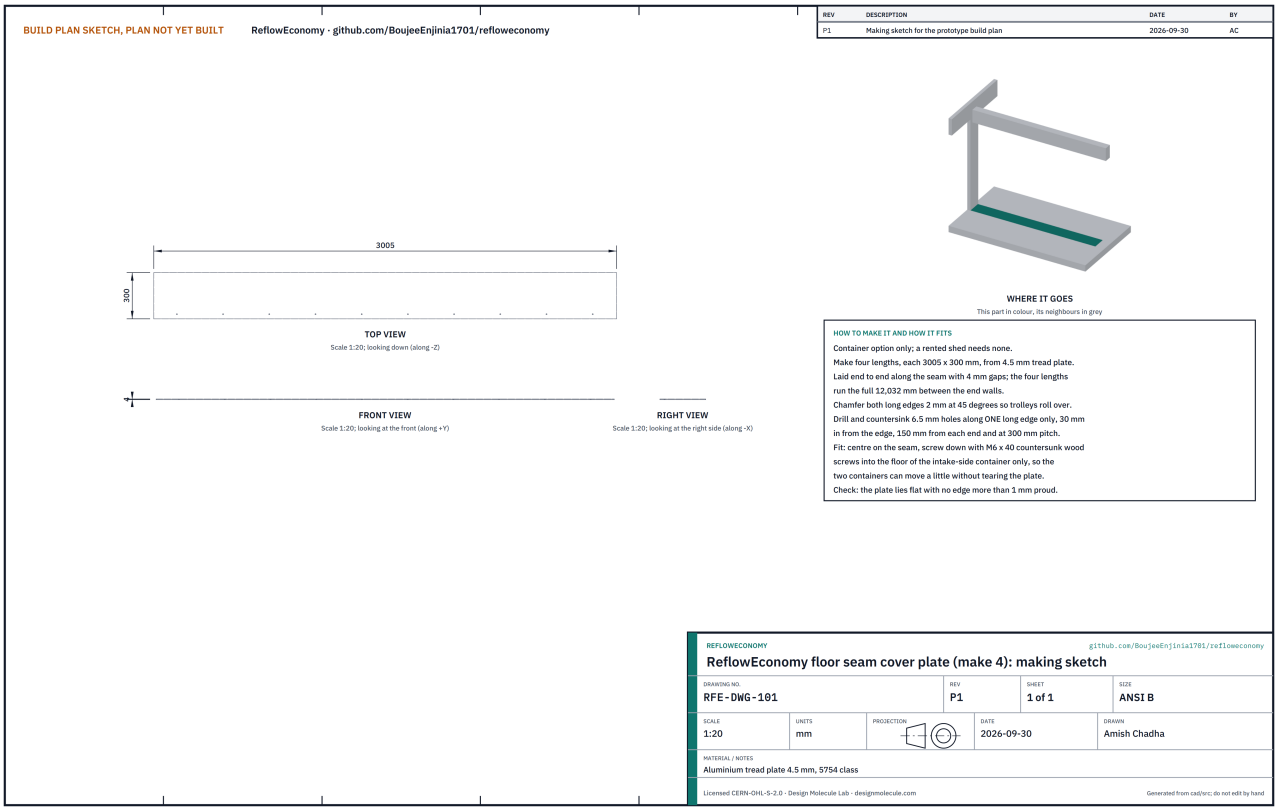


Figure 3. Floor seam cover plate making sketch (RFE-DWG-101).

What it is and what it is made from. A strip of tread plate that covers the joint between the two container floors down the middle of the aisle. Aluminium tread plate 4.5 mm thick, 5754 class, in four lengths of 3,005 x 300.

How to make it.

1. Cut four lengths 3,005 x 300 and square the ends.
2. Chamfer both long edges 2 mm at 45° so trolley wheels roll over them.
3. Along one long edge only, 30 from the edge, drill and countersink 6.5 mm holes: the first 150 from one end, then every 300 (ten holes per length).
4. Deburr every edge and hole.

How it fits the parts next to it.

Joint 1: seam beam and floor seam plate at the intake end

Container option only. The beam sits on the corner posts; its section and fixings are the structural engineer's

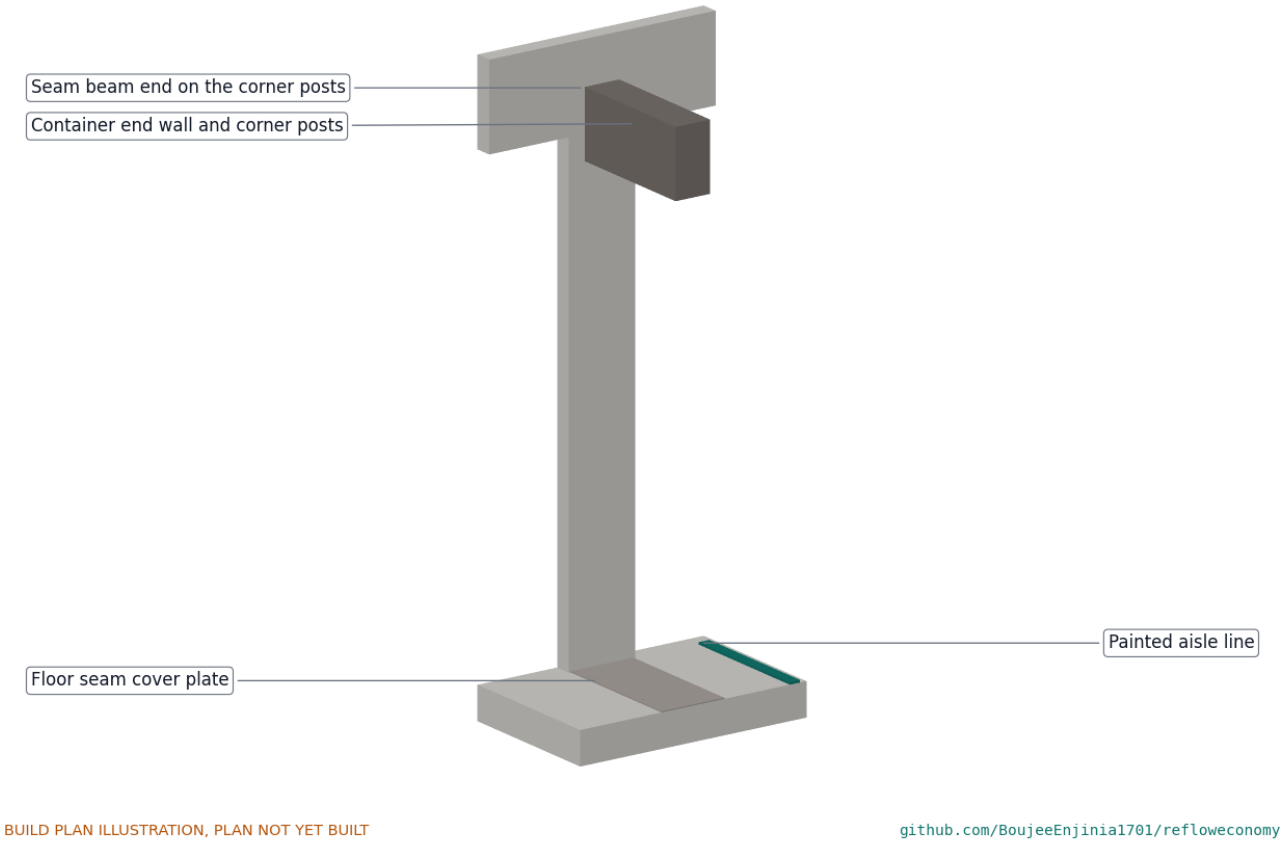


Figure 4. The plate is centred on the seam; the beam above it sits on the corner posts.

The four lengths lie end to end along the seam with 4 mm gaps, together running the 12,032 between the end walls. The drilled edge goes over the intake-side container and is screwed into its floor with M6 x 40 countersunk wood screws; the other edge rests on the other container's floor, unfixed, so the two containers can move a little without tearing the plate.

Check before moving on. No edge stands more than 1 mm proud of the floor.

3.3 Floor markings

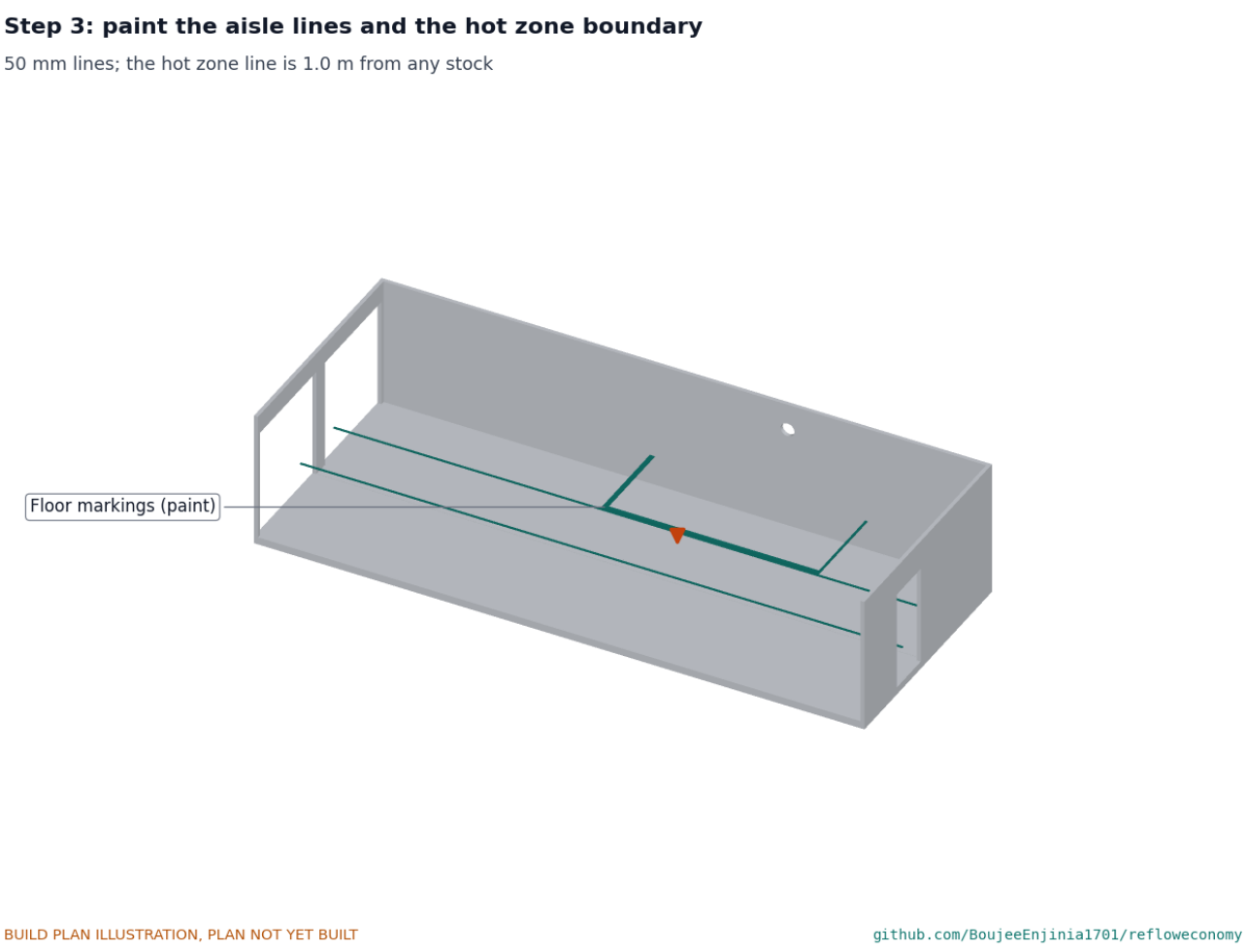


Figure 5. The painted aisle edges and the hot zone boundary.

Two 50 mm lines mark the aisle edges, 1,650 to 1,700 and 2,900 to 2,950 from the front wall, along the whole floor. The hot zone boundary runs from 5,400 to 9,750 along the line: a 100 mm band across the front of the hot zone, 2,960 to 3,060 from the front wall, and legs back to the back wall at each end (the far leg is 50 wide and runs along the outside of the press booth). Use a floor paint for plywood floors; nothing is stored inside the boundary, and stock stays at least 1.0 m outside it.

3.4 Safety station (bought)

A PPE and first aid cabinet 800 x 450 x 1,900 standing against the back wall at 2,500 to 3,300 along, held by two wall brackets; an eyewash fitted to the backboard (section 3.5); and three fire extinguishers on floor stands, two at the hot zone entrance (4,800 and 5,300 along, against the back wall) and one at the product end (10,600 along, beside the aisle). It goes in first, before any power or hot work (step 4).

3.5 Backboard

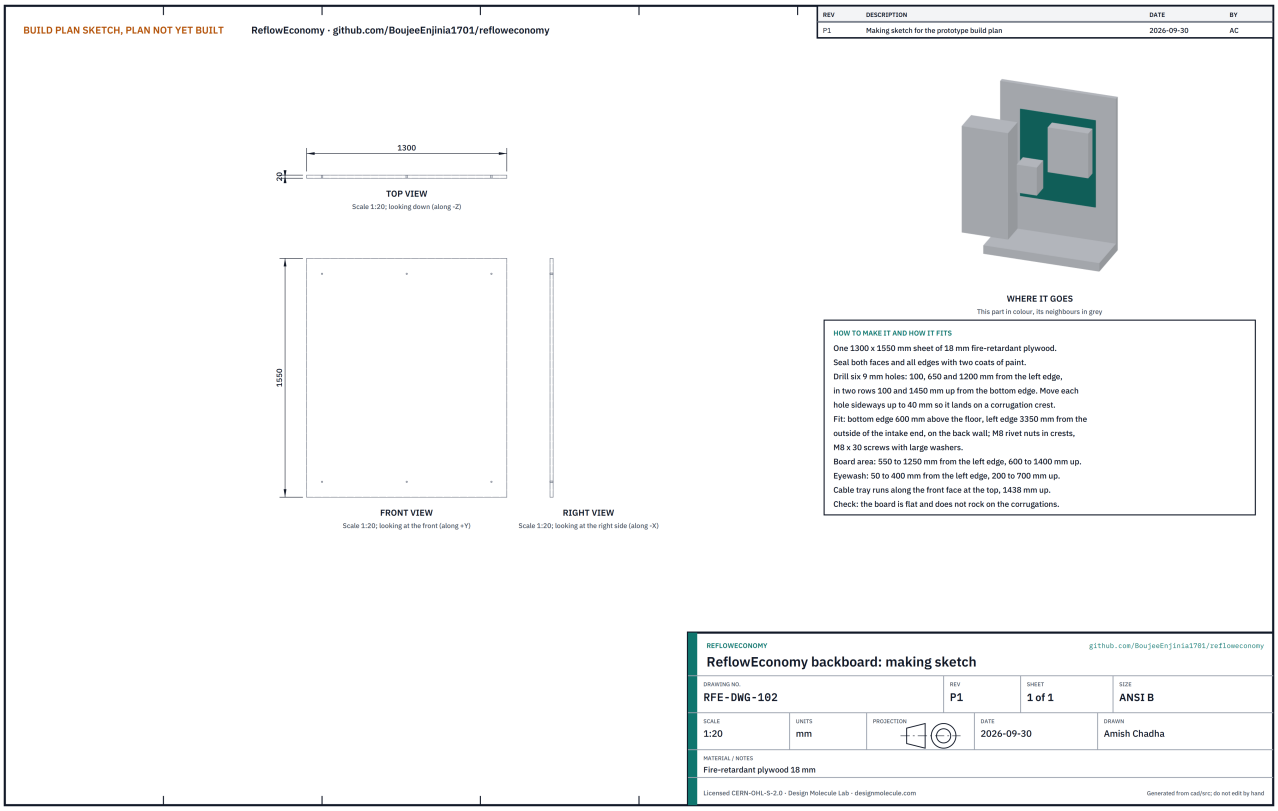


Figure 6. Backboard making sketch (RFE-DWG-102).

What it is and what it is made from. A flat board on the corrugated back wall that the electrical board and the eyewash screw to. One sheet of 18 mm fire-retardant plywood, cut to 1,300 x 1,550.

How to make it.

1. Cut the sheet to 1,300 wide x 1,550 tall and seal both faces and every edge with two coats of paint.
2. Drill six 9 mm holes: 100, 650 and 1,200 from the left edge, in two rows 100 and 1,450 up from the bottom edge.
3. Hold it on the wall and move any hole sideways by up to 40 so it lands on a corrugation crest; fit an M8 rivet nut in the wall behind each hole.

How it fits the parts next to it.

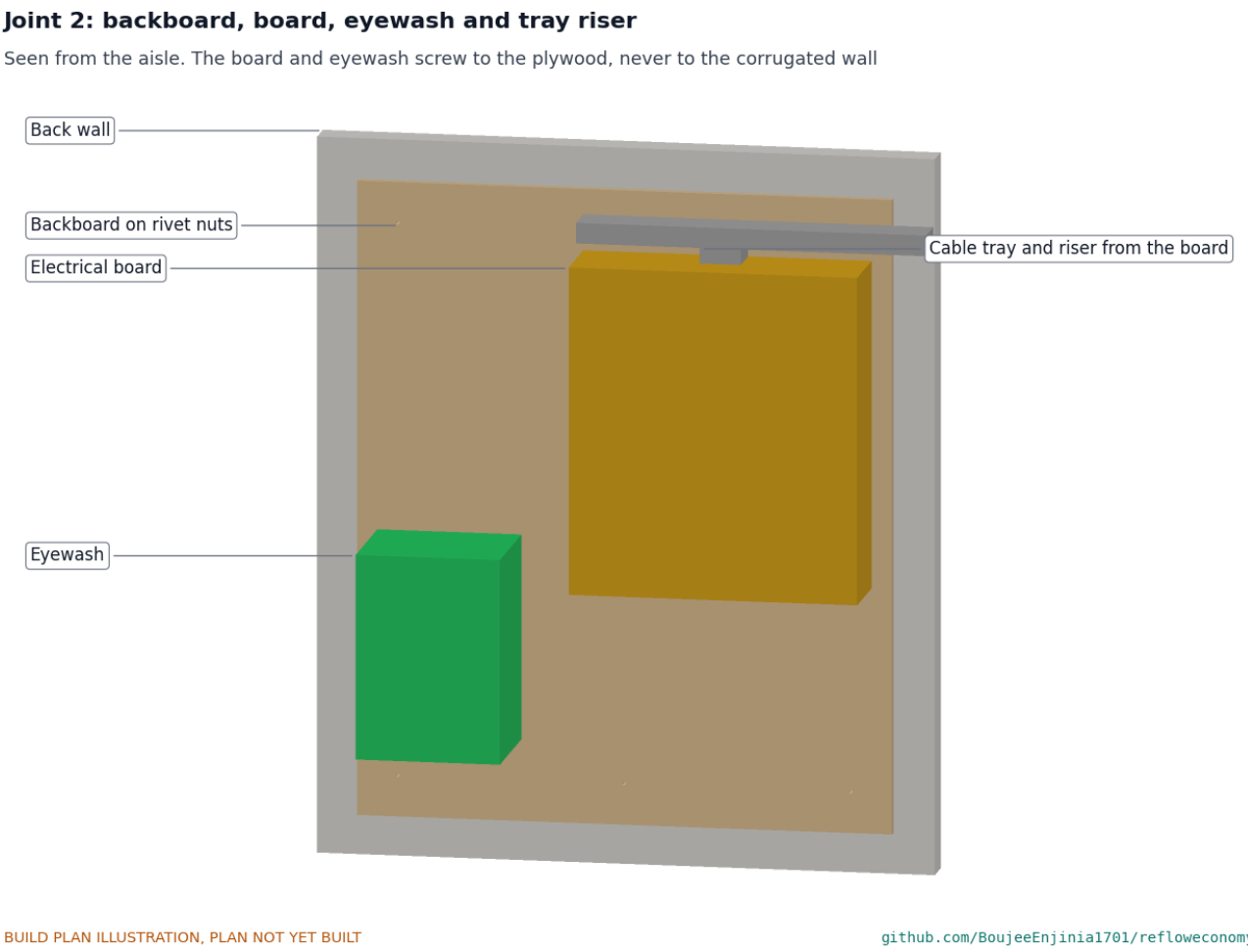


Figure 7. The board and the eyewash screw to the plywood, never to the corrugated wall.

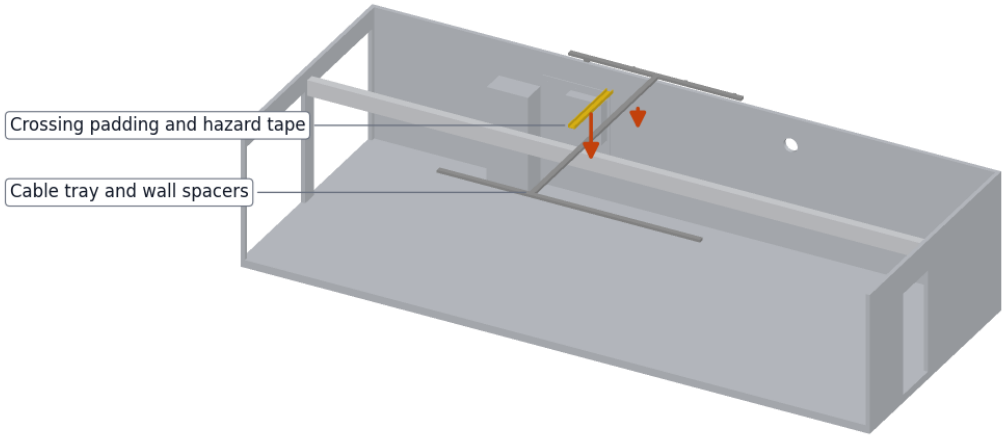
The backboard sits on the back wall from 3,350 to 4,650 along and from 600 to 2,150 above the floor, held by six M8 x 30 screws with large washers into the rivet nuts. The electrical board covers 550 to 1,250 from the board's left edge and 600 to 1,400 up from its bottom edge; the eyewash covers 50 to 400 across and 200 to 700 up. The cable tray runs across the front of the board's top, 1,438 up.

Check before moving on. The board does not rock on the corrugations; every screw is tight.

3.6 Electrical board and cable tray (licensed electrician)

Step 7: cable tray

Back run, crossing under the seam beam with taped padding over the aisle, front run



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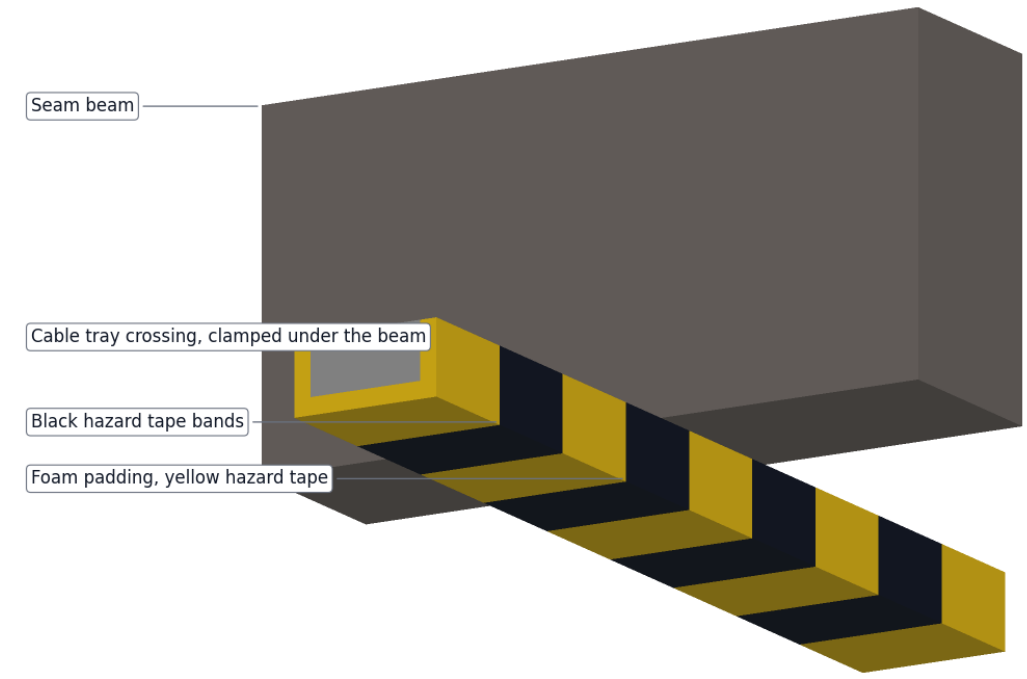
Figure 8. The tray runs along the back wall, crosses the aisle once under the seam beam and runs along the front wall.

What to buy and fit. A distribution board for a 230 V, 40 A single-phase supply with 30 mA residual current devices on every circuit, circuit breakers, an emergency stop circuit, the heater interlock contactor that locks the press heaters out while the shredder or extruder runs, and an energy meter. About 20 m of 100 x 50 perforated galvanised cable tray with three wall spacers, two beam clamps and a short riser trunking from the board top. For the aisle crossing: 1.4 m of 15 mm closed-cell foam edge guard and a roll of yellow and black hazard tape.

How it fits. The tray's underside is 2,038 above the floor everywhere. The back run sits on the backboard and on 20 mm spacers riveted to the wall at about 5,100, 6,100 and 7,050 along, from 3,900 to the end wall of the press booth, where the cables pass through a gland. The crossing runs across the aisle at 5,550 to 5,650 along, clamped under the seam beam. Over the aisle, from 1,600 to 3,000 from the front wall, the foam is fitted round its underside and sides and wrapped in hazard tape in 100 mm yellow and black bands, so a head that touches it meets something soft and seen. The front run is screwed to the front wall from 3,800 to 8,900 along, over the washing tanks, the shredder enclosure and the drying rack.

Joint 3: padded cable tray crossing under the seam beam

Two beam clamps hold the tray at 2,038 mm; the taped padding leaves 2,023 mm of headroom



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Figure 9. Two beam clamps carry the crossing; under its taped padding there is 2.02 m of headroom over the aisle.

Check before moving on. The electrician's installation certificate and residual current device trip tests are done (stop S3).

3.7 Extruder (built to its published design)

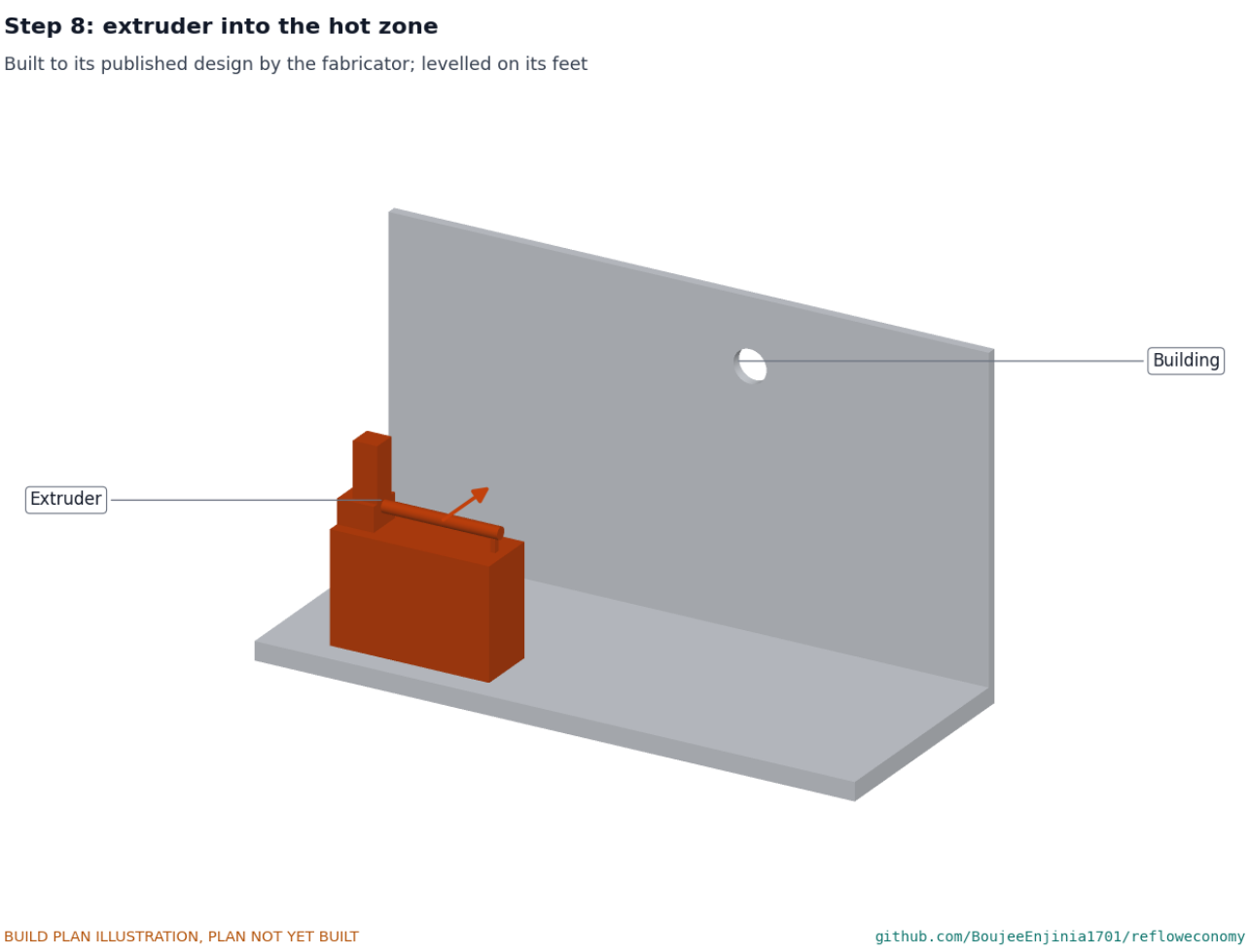


Figure 10. The extruder stands in the hot zone against the back wall.

What to buy or have built. An open-design extruder for beams and profiles, for HDPE and PP only, on a frame 1,300 long x 500 deep with a flat top 900 above the floor: a 1.5 kW motor on a single-phase input drive and 2.0 kW of barrel heaters, the hopper at the drive end. It stands at 5,700 to 7,000 along, 4,000 to 4,500 from the front wall, barrel along the line.

Check before moving on. The frame top is flat and level, with room for six M8 bolts round the barrel (section 3.8).

3.8 Hood B (extruder barrel enclosure)

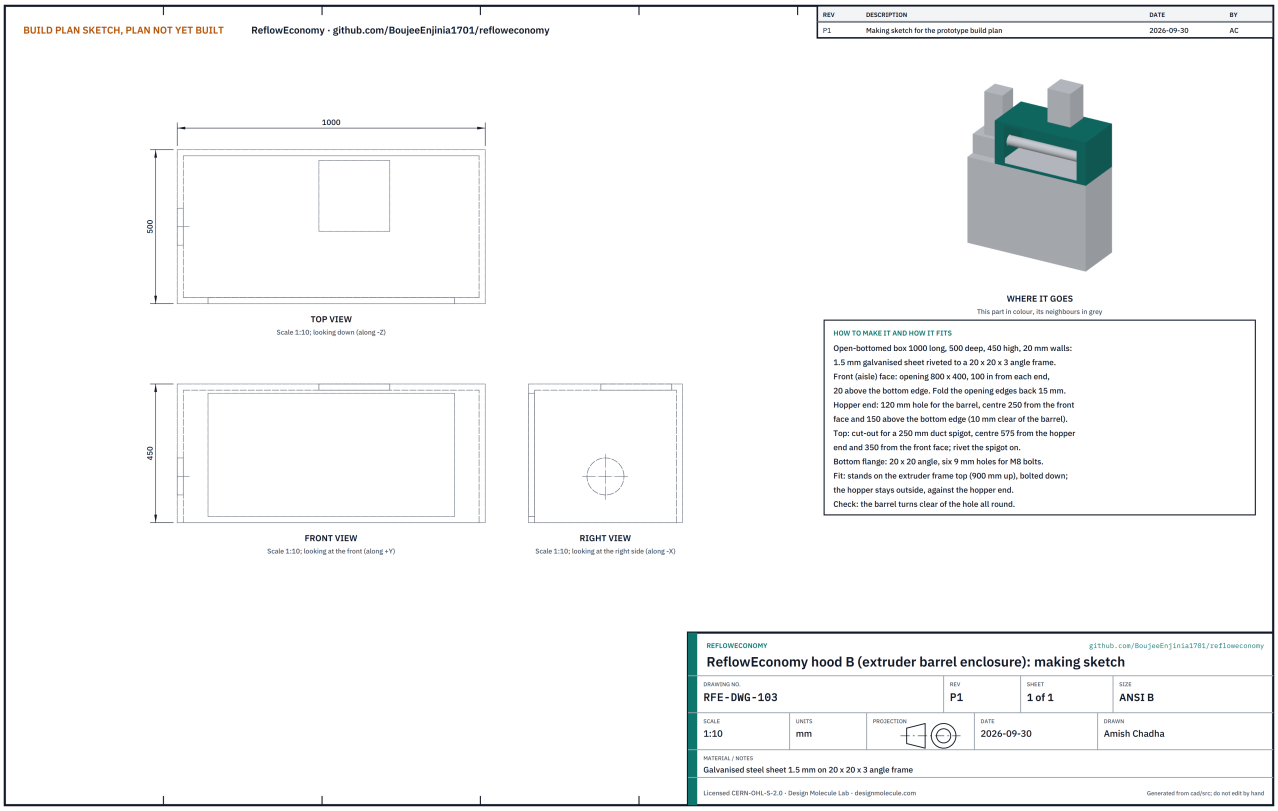


Figure 11. Hood B making sketch (RFE-DWG-103).

What it is and what it is made from. An open-bottomed sheet box over the hot barrel that the duct extracts. Galvanised steel sheet 1.5 mm riveted to a frame of 20 x 20 x 3 angle; the walls are drawn 20 thick.

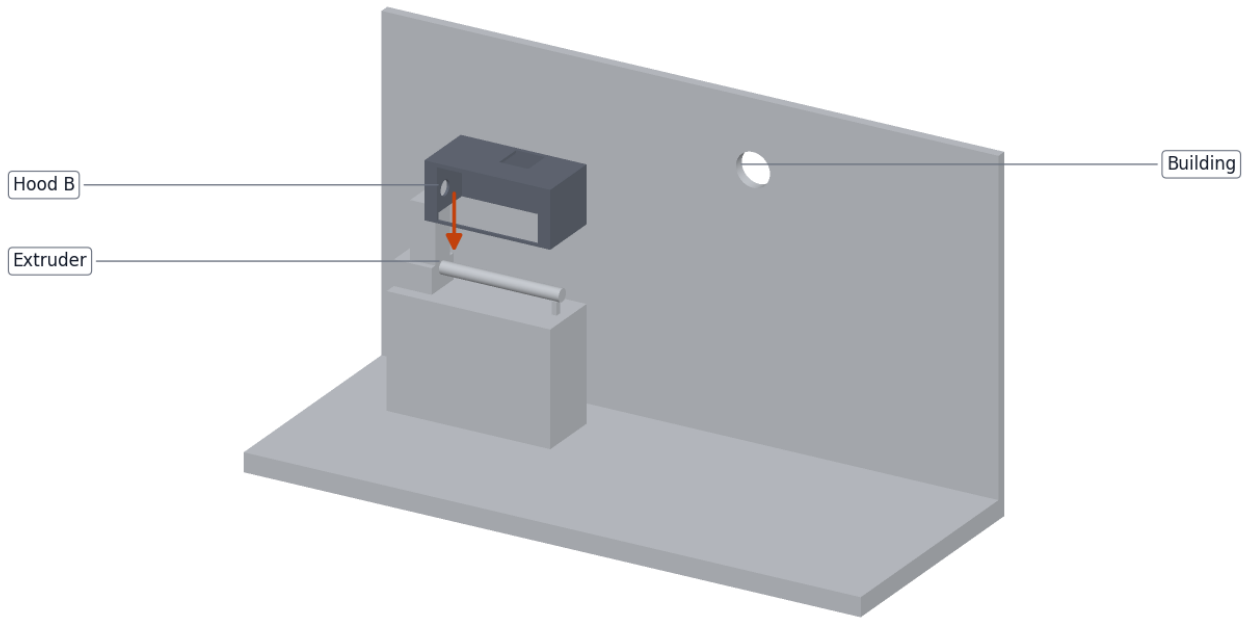
How to make it.

1. Make the frame 1,000 long, 500 deep and 450 high from 20 x 20 x 3 angle, with a bottom flange of the same angle all round.
2. Front (aisle) face: cut an opening 800 wide x 400 tall, 100 in from each end and 20 up from the bottom edge; fold its edges back 15.
3. Hopper end: cut a 120 mm hole for the barrel, centred 250 from the front face and 150 up from the bottom edge.
4. Top: cut the hole for a 250 mm round duct spigot, centred 575 from the hopper end and 350 from the front face; rivet the spigot on.
5. Rivet the sheet to the frame. Drill six 9 mm holes in the bottom flange to match holes drilled in the extruder frame top.

How it fits the parts next to it.

Step 9: hood B onto the extruder frame

Six M8 bolts through its bottom flange; the barrel passes through the end hole



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Figure 12. Hood B goes over the barrel and down onto the frame.

Joint 4: hood B on the extruder frame (cut along the barrel)

The barrel runs through a 120 mm hole in the hood end; the open face is toward the aisle

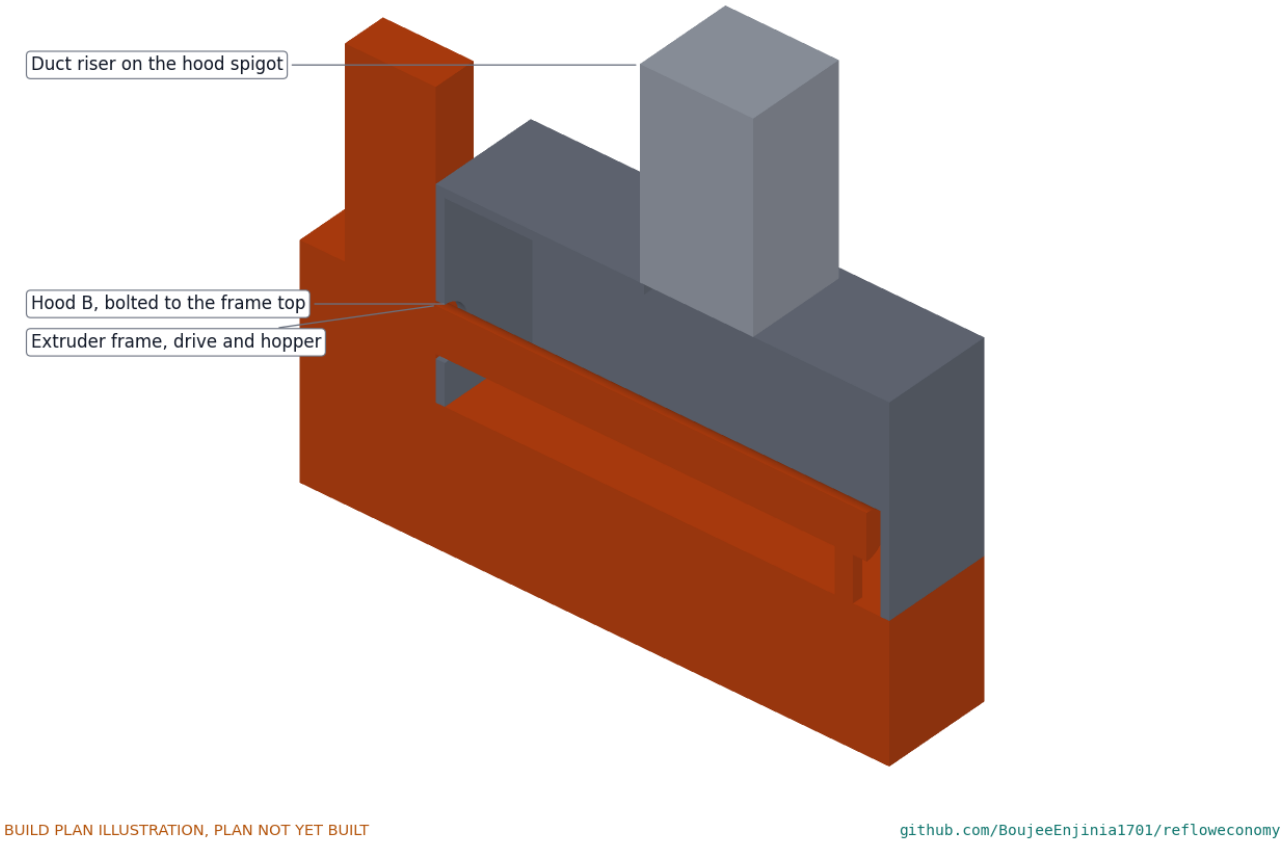


Figure 13. Cut along the barrel: the hood stands on the frame top; the barrel runs through the end hole with 10 mm clear all round.

The bottom flange sits on the extruder frame top, 900 above the floor, held by six M8 bolts. The hopper stays outside, against the hopper end. The duct riser stands on the top spigot.

Check before moving on. The barrel turns without touching the hole; the open face faces the aisle.

3.9 Sheet press and cooling press (built to their published designs)

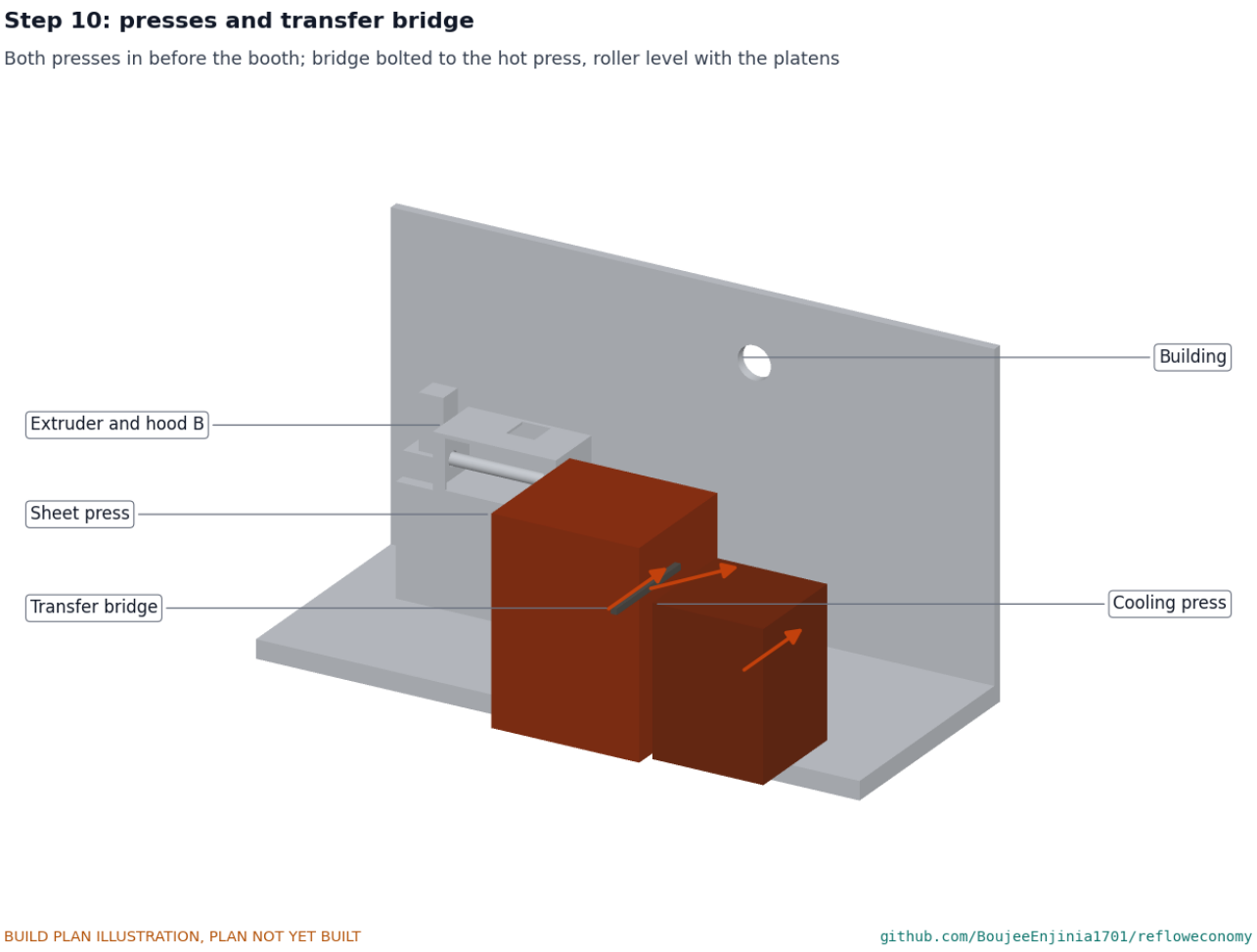


Figure 14. Both presses go in before the booth is built round them.

What to buy or have built. A 1 x 1 m heated sheet press with 5.0 kW single-phase heaters and insulated platens, on a frame about 1,200 x 1,100 (the derated single-phase variant of the published press), standing at 7,400 to 8,600 along, 3,500 to 4,600 from the front wall; and an unheated cooling press about 900 x 900, standing at 8,650 to 9,550 along, 3,600 to 4,500 from the front wall. Both platens are at the same working height, taken here as 950 above the floor.

3.10 Transfer bridge

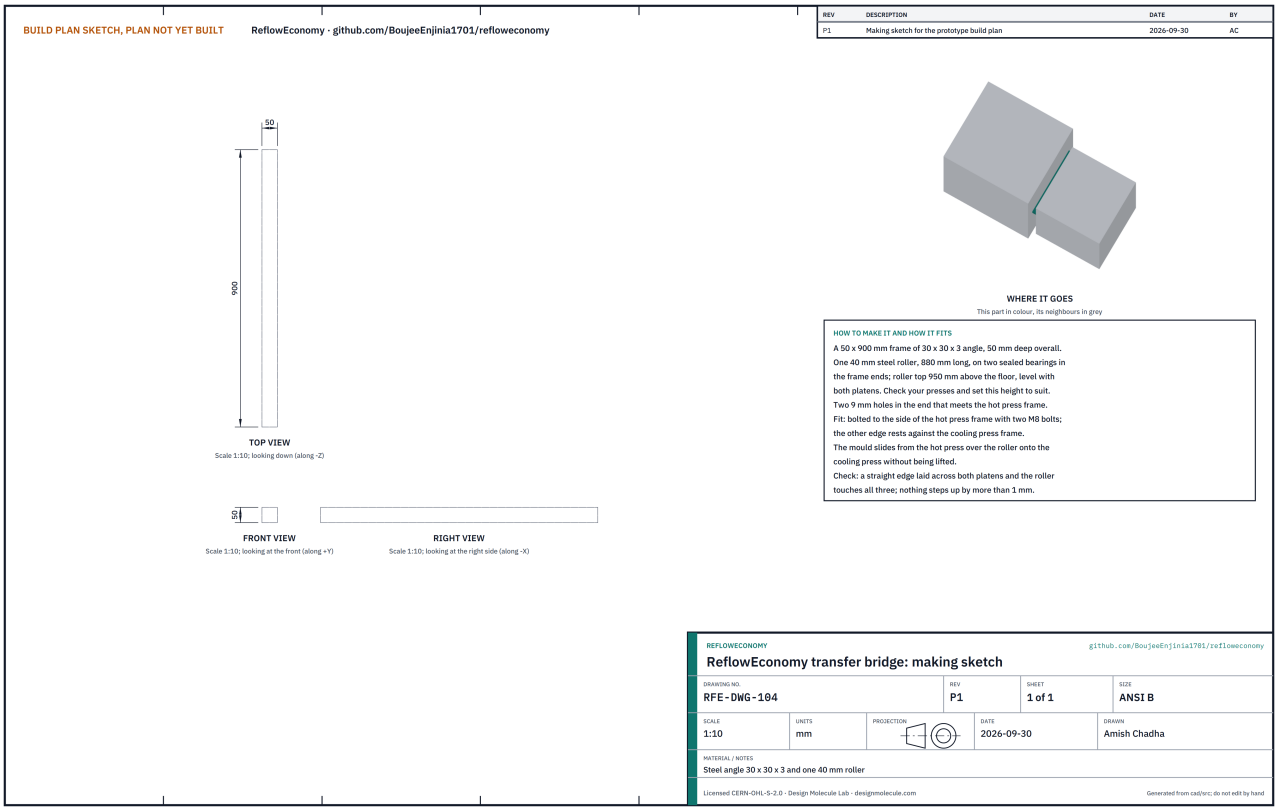


Figure 15. Transfer bridge making sketch (RFE-DWG-104).

What it is and what it is made from. A short roller bridge across the 50 mm gap between the presses, so the hot mould slides from the hot press onto the cooling press without being lifted. Steel angle 30 x 30 x 3 and one 40 mm steel roller on sealed bearings.

How to make it.

1. Weld a frame 50 wide x 900 long x 50 deep from 30 x 30 x 3 angle.
2. Fit one 40 mm roller, 880 long, on two sealed bearings in the frame ends, its top 950 above the floor (level with both platens; set this to suit the presses you have).
3. Drill two 9 mm holes in the side that meets the hot press frame.

How it fits the parts next to it.

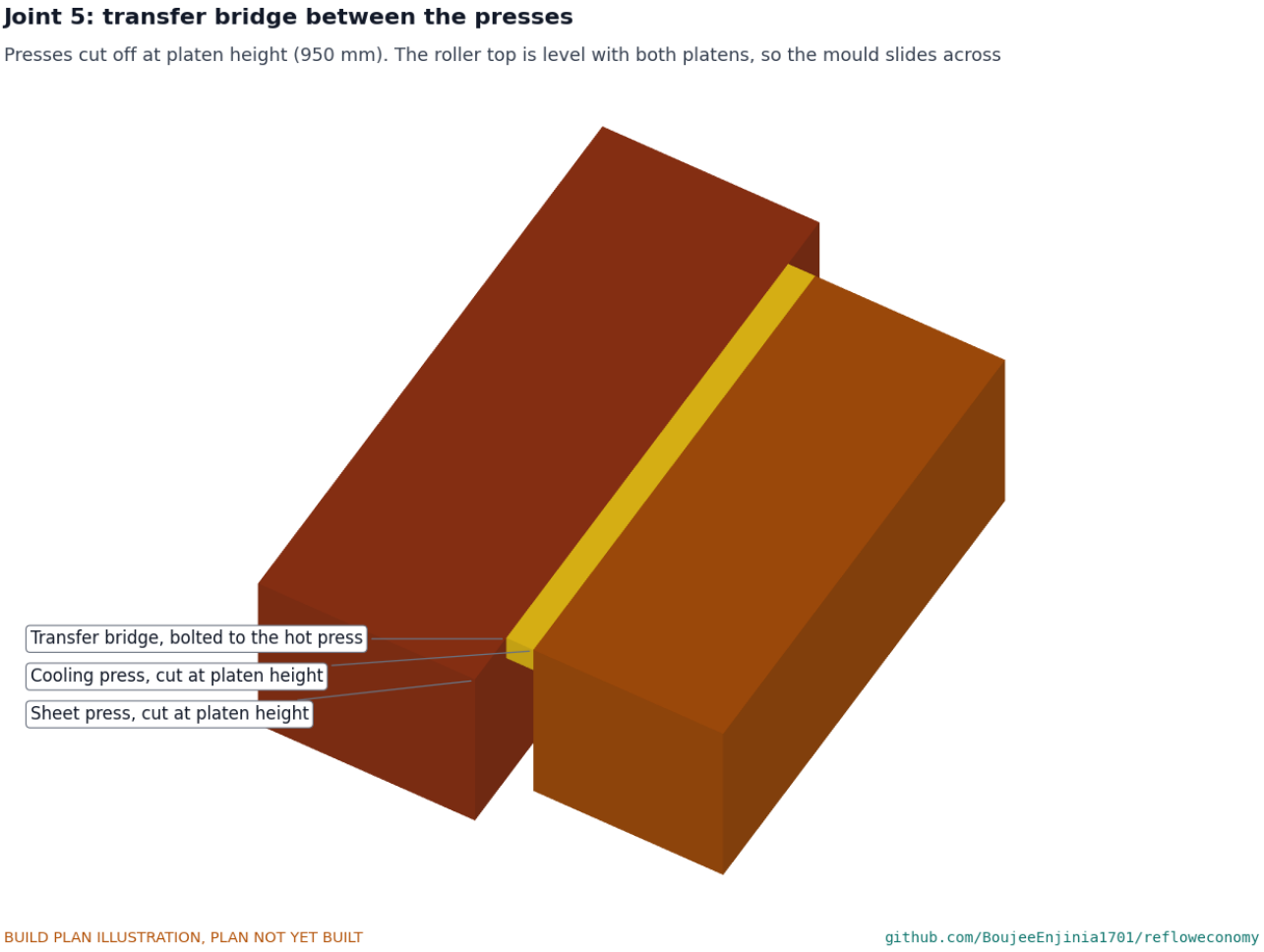


Figure 16. Presses cut off at platen height: the roller top is level with both platens.

One side bolts to the hot press frame with two M8 bolts; the other rests against the cooling press frame.

Check before moving on. A straight edge laid across both platens and the roller touches all three; nothing steps up by more than 1 mm.

3.11 Press booth (hood A)

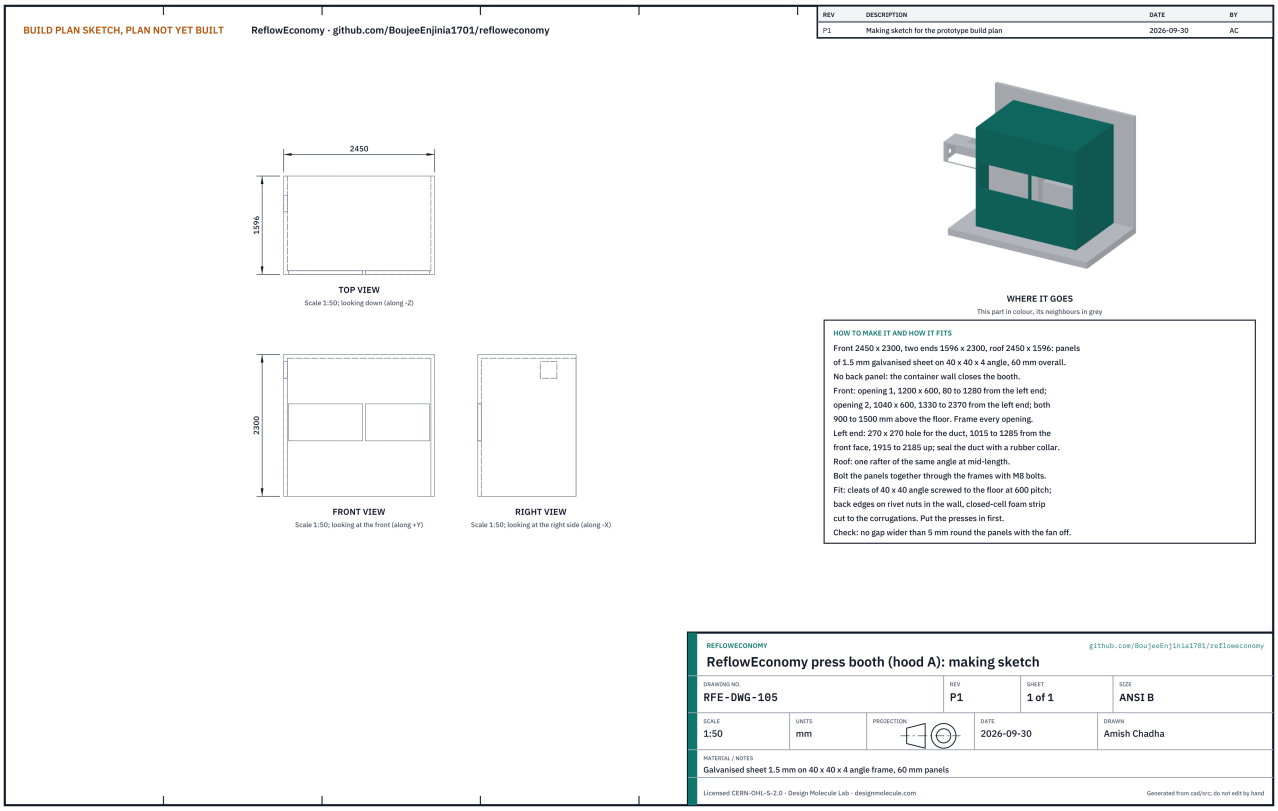


Figure 17. Press booth making sketch (RFE-DWG-105).

What it is and what it is made from. An enclosure round both presses, open at the back where the container wall closes it, extracted by the duct. Panels of 1.5 mm galvanised sheet on 40 x 40 x 4 angle frames, 60 overall.

How to make it.

1. Make a front panel 2,450 x 2,300, two end panels 1,596 x 2,300 and a roof 2,450 x 1,596, each a welded frame of 40 x 40 x 4 angle with sheet riveted to its outer face; give the roof one rafter of the same angle at mid-length.
2. Front panel: frame two openings, both 900 to 1,500 above the floor: opening 1, 1,200 wide, from 80 to 1,280 from the left end; opening 2, 1,040 wide, from 1,330 to 2,370.
3. Left end panel: cut and frame a 270 x 270 hole for the duct, 1,015 to 1,285 from the front face and 1,915 to 2,185 above the floor.
4. Drill the frames for M8 bolts at 400 pitch where the panels meet.

How it fits the parts next to it.

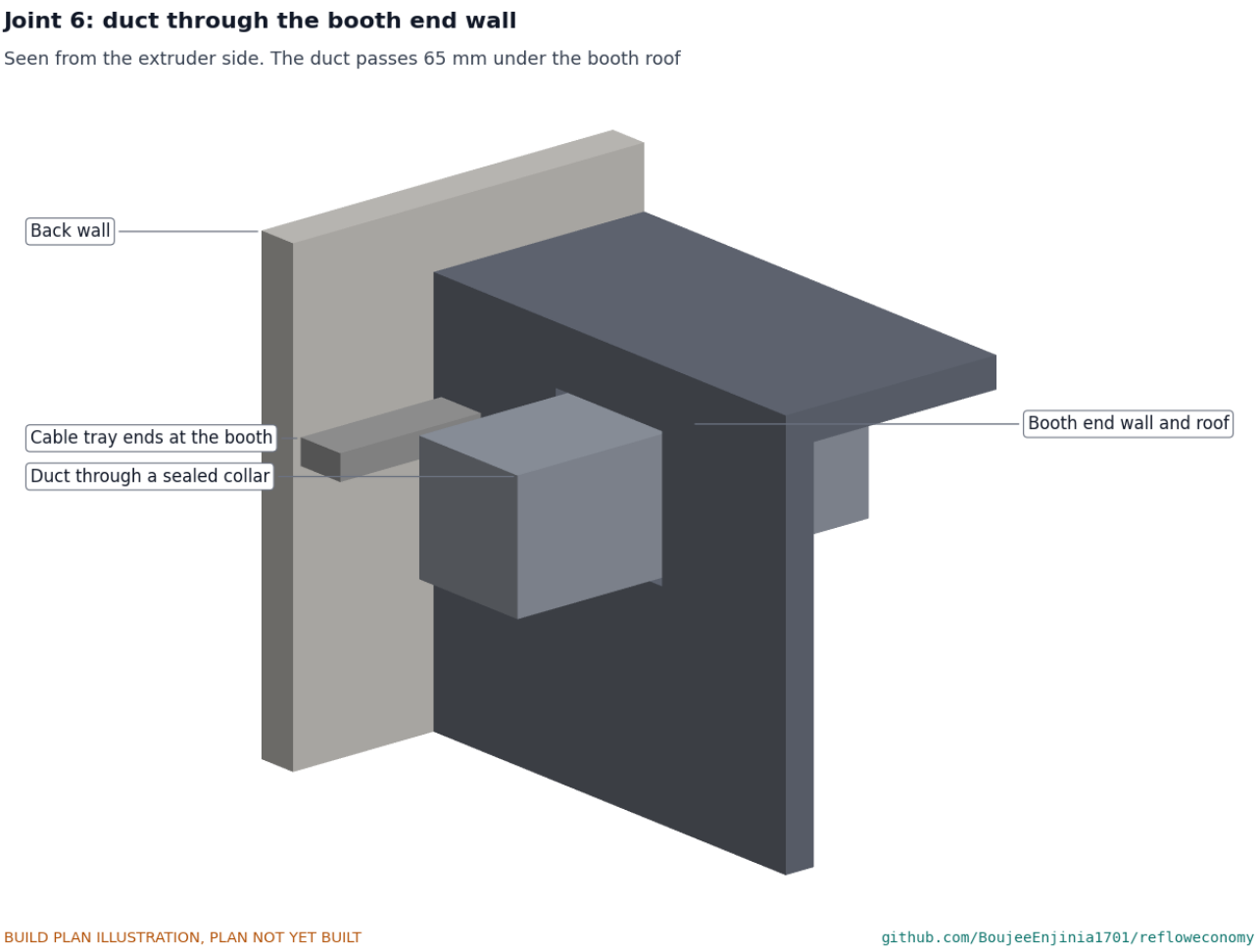


Figure 18. Seen from the extruder side: the duct passes through a sealed collar in the end wall, 65 mm under the roof.

The booth stands from 7,250 to 9,700 along and from 3,200 from the front wall back to the back wall, 2,300 high, round the two presses. The panels bolt together through their frames. The bottom frames are held to the floor by 40 x 40 angle cleats screwed down at 600 pitch; the back edges are screwed to rivet nuts in the wall, over a closed-cell foam strip cut to the corrugations. A rubber collar seals the duct in its hole.

Check before moving on. With the fan off, no gap round any panel is wider than 5 mm.

3.12 Sliding sash and rails

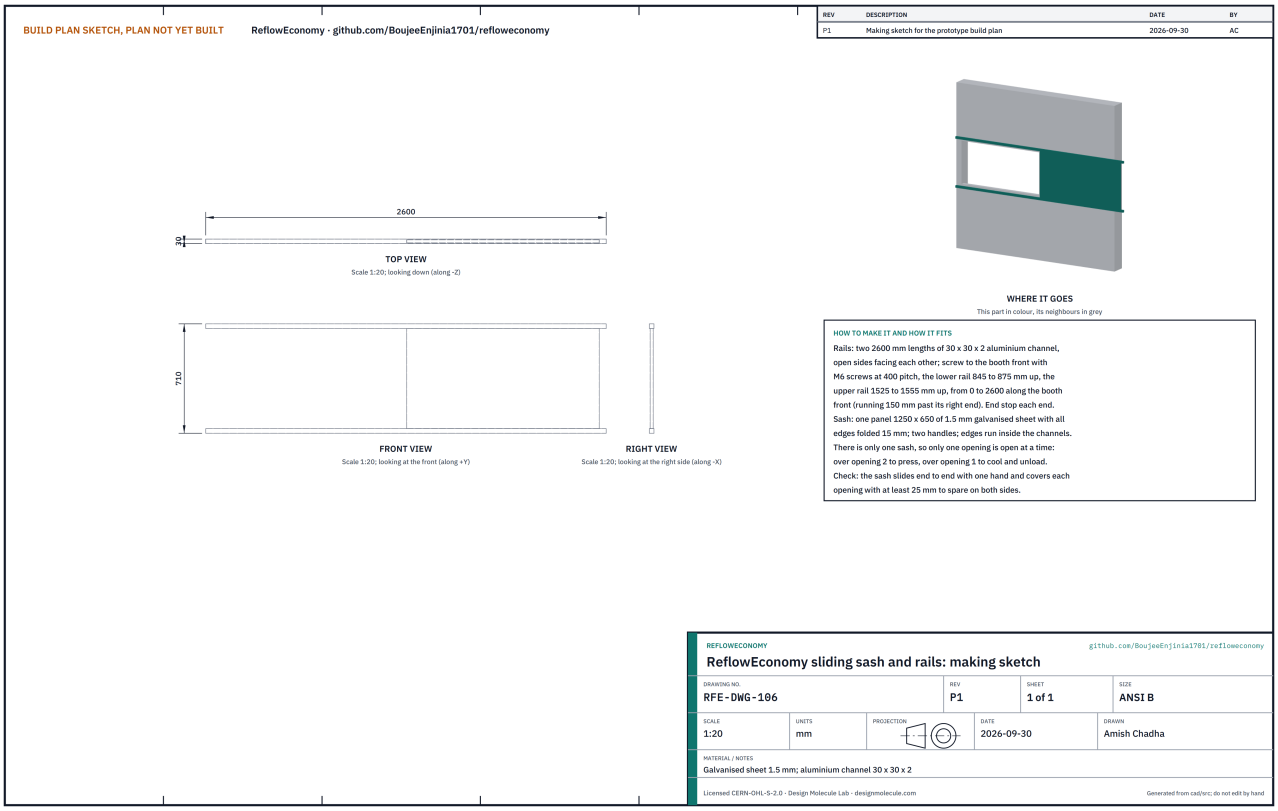


Figure 19. Sliding sash and rails making sketch (RFE-DWG-106).

What it is and what it is made from. One panel that slides along the booth front and always covers one of the two openings, so only one is ever open. Rails of 30 x 30 x 2 aluminium channel; the panel is 1.5 mm galvanised sheet.

How to make it.

1. Cut two rails 2,600 long. Drill 6.5 mm holes at 400 pitch through the channel back.
2. Cut the panel 1,250 x 650 with 15 mm folded edges on all sides, and fit two handles on its front.
3. Make two end stops from offcuts.

How it fits the parts next to it.

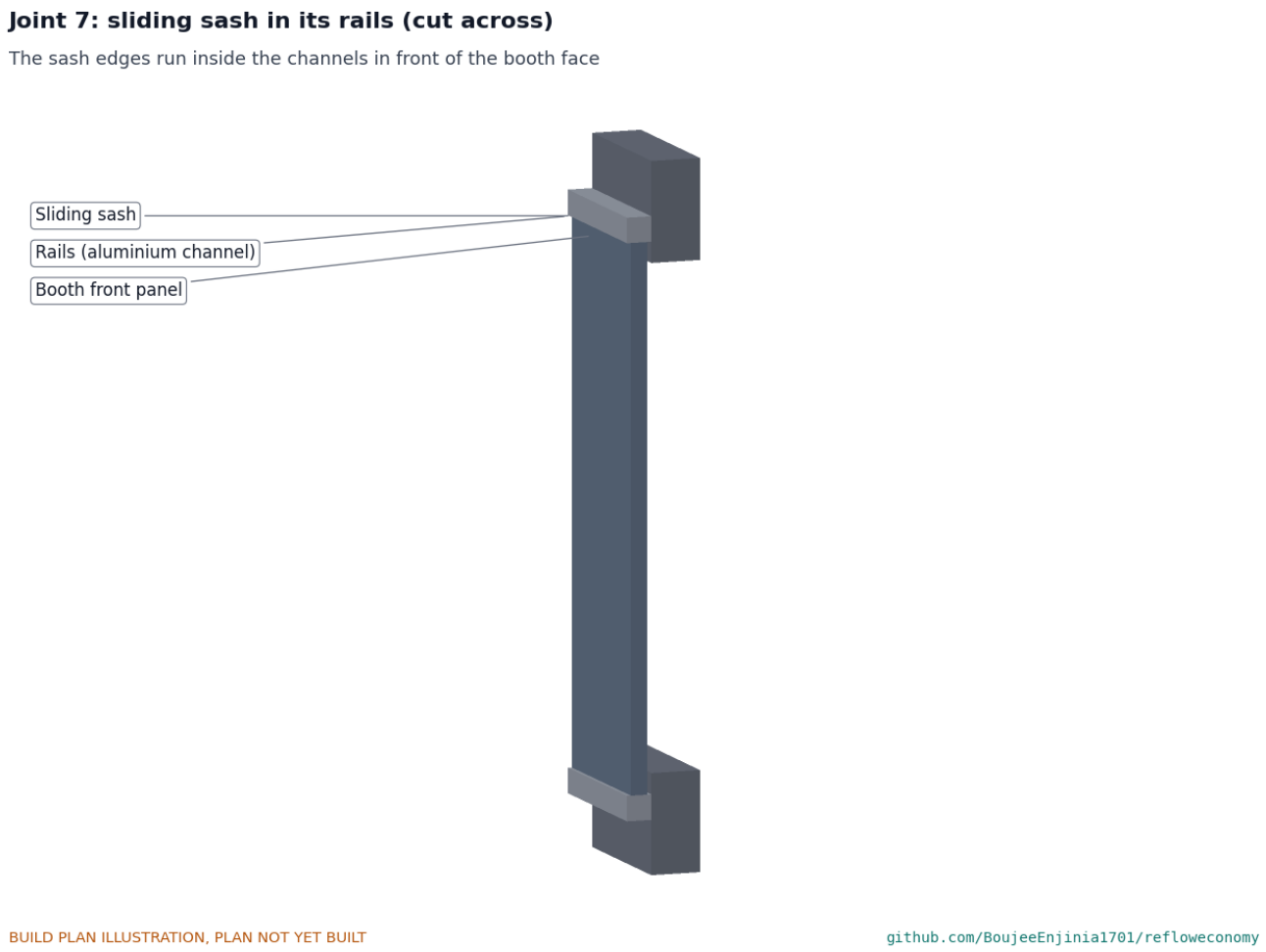


Figure 20. Cut across: the sash's top and bottom edges run inside the channels in front of the booth face.

The rails screw to the booth front with M6 screws, open sides facing each other: the lower one 845 to 875 above the floor, the upper one 1,525 to 1,555, both running from the booth's left end to 150 past its right end. The sash covers opening 2 while a sheet is pressed (opening 1 open) and opening 1 while the mould is moved to the cooling press and unloaded (opening 2 open).

Check before moving on. The sash slides end to end with one hand and covers each opening with at least 25 to spare on both sides.

3.13 Duct, booth take-off and duct wall brackets

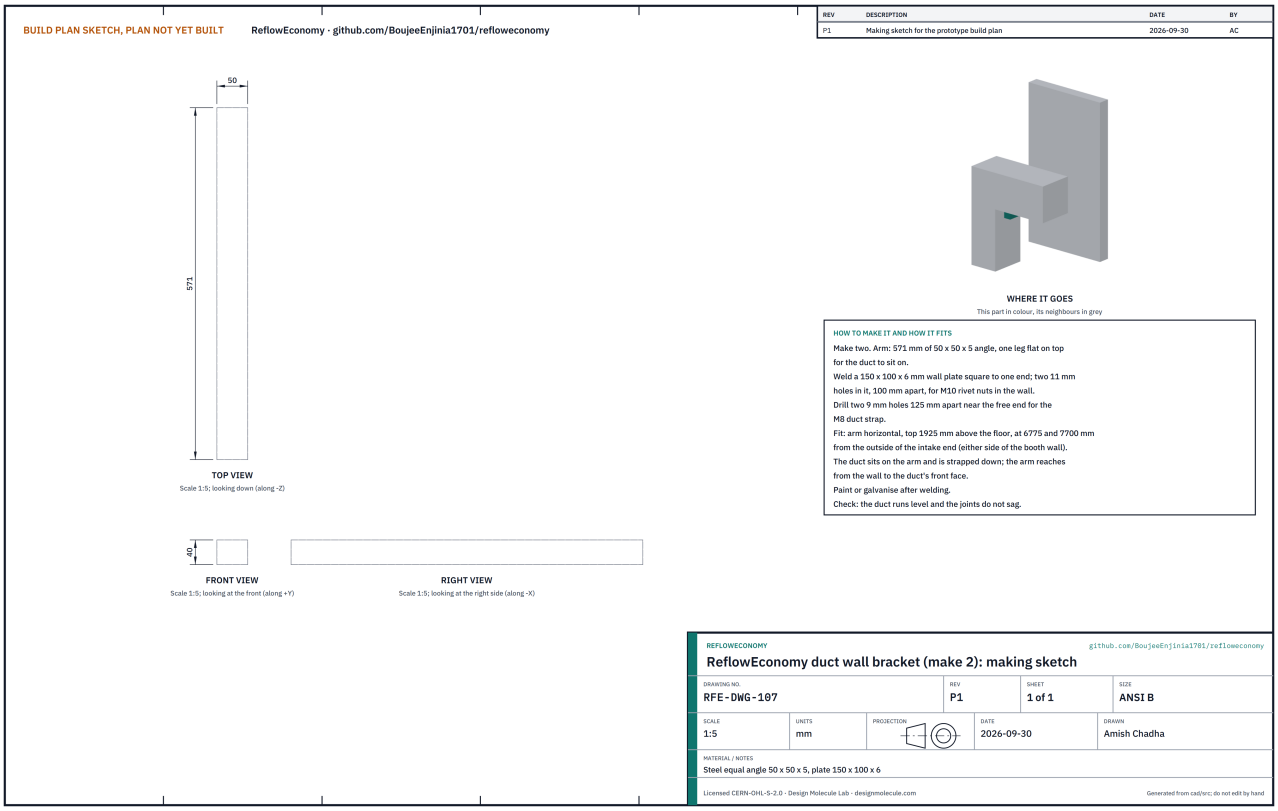


Figure 21. Duct wall bracket making sketch (RFE-DWG-107).

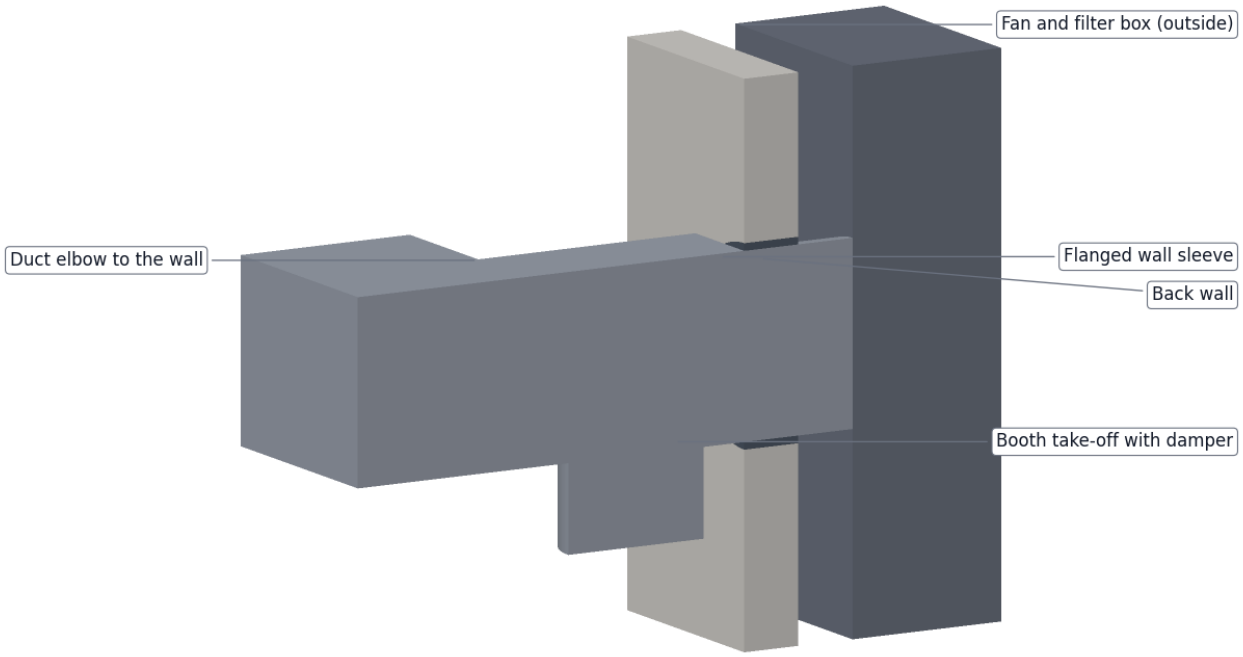
What to buy. 250 mm galvanised spiral duct and fittings: a riser from hood B, a straight run, a two-way elbow junction, a booth take-off with a balancing damper, and a flanged wall sleeve for the back wall hole.

What to make: two wall brackets. Each is a 571 long arm of 50 x 50 x 5 equal angle, one leg flat on top, with a 150 x 100 x 6 wall plate welded square to one end (two 11 mm holes 100 apart) and two 9 mm holes 125 apart near the free end for an M8 duct strap. Paint or galvanise after welding.

How it fits the parts next to it.

Joint 8: duct elbow, booth take-off and wall sleeve (cut through the duct)

Inside the booth: the take-off draws the booth air; the extruder branch joins at the elbow



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Figure 22. Cut through the duct: inside the booth the take-off draws the booth air, and the hood B branch joins at the elbow.

The riser stands on the hood B spigot and rises to a run centred 2,050 above the floor and 4,350 from the front wall. The run goes along the back row, through the booth end wall, to the elbow at 8,150 along, which turns it to the back wall and through the sleeve to the fan box. The booth take-off points down from the elbow. The brackets bolt to M10 rivet nuts in the back wall, arm tops 1,925 above the floor, at 6,775 and 7,700 along (one each side of the booth end wall); the duct sits on them and is strapped down.

Check before moving on. The run is level and every joint is sealed with duct tape over a sealant bead.

3.14 Fan stand, fan and filter box, and stack

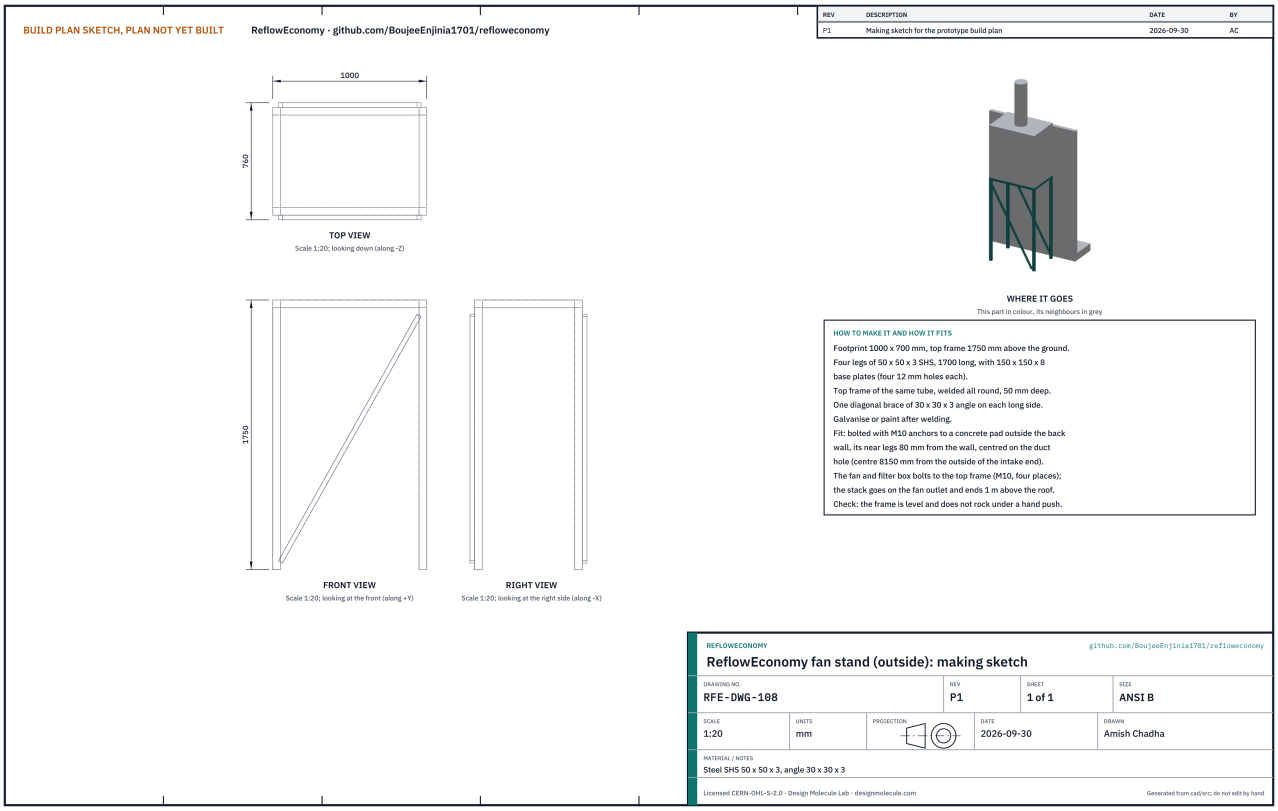


Figure 23. Fan stand making sketch (RFE-DWG-108).

What it is and what it is made from. A braced steel stand outside the back wall that carries the fan and filter box at duct height. Steel square hollow section 50 x 50 x 3 and angle 30 x 30 x 3.

How to make it.

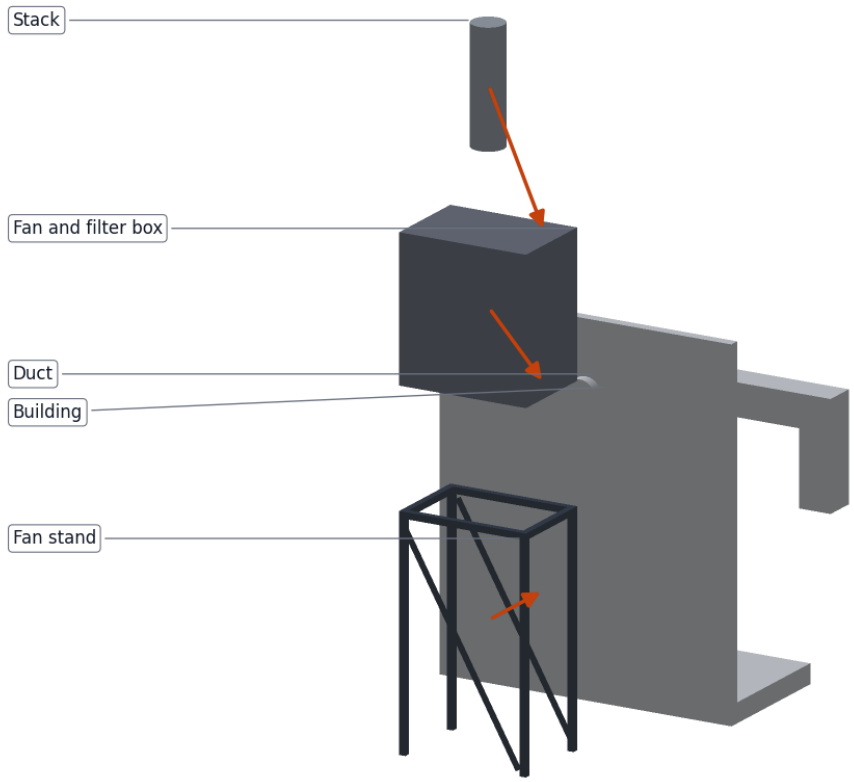
1. Cut four legs of 50 x 50 x 3 tube, 1,700 long, and weld a 150 x 150 x 8 base plate to each with four 12 mm holes.
2. Weld a top frame of the same tube, 1,000 x 700 outside, onto the legs.
3. Weld one diagonal brace of 30 x 30 x 3 angle across each long side, from low on one leg to high on the other.
4. Drill four 11 mm holes in the top frame to match the fan box feet. Galvanise or paint.

What to buy. A 1.5 kW centrifugal fan in a box with a G4 prefilter, an F7 bag filter and an impregnated activated carbon stage, about 1,000 x 700 x 1,100; a 250 mm stack that ends 1 m above the roof, with a rain cap that does not deflect the plume downward.

How it fits the parts next to it.

Step 14: fan stand, fan and filter box, stack (outside)

Seen from behind. Stand anchored to a pad; fan box bolted on; stack 1 m above the roof



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Figure 24. Seen from behind the factory.

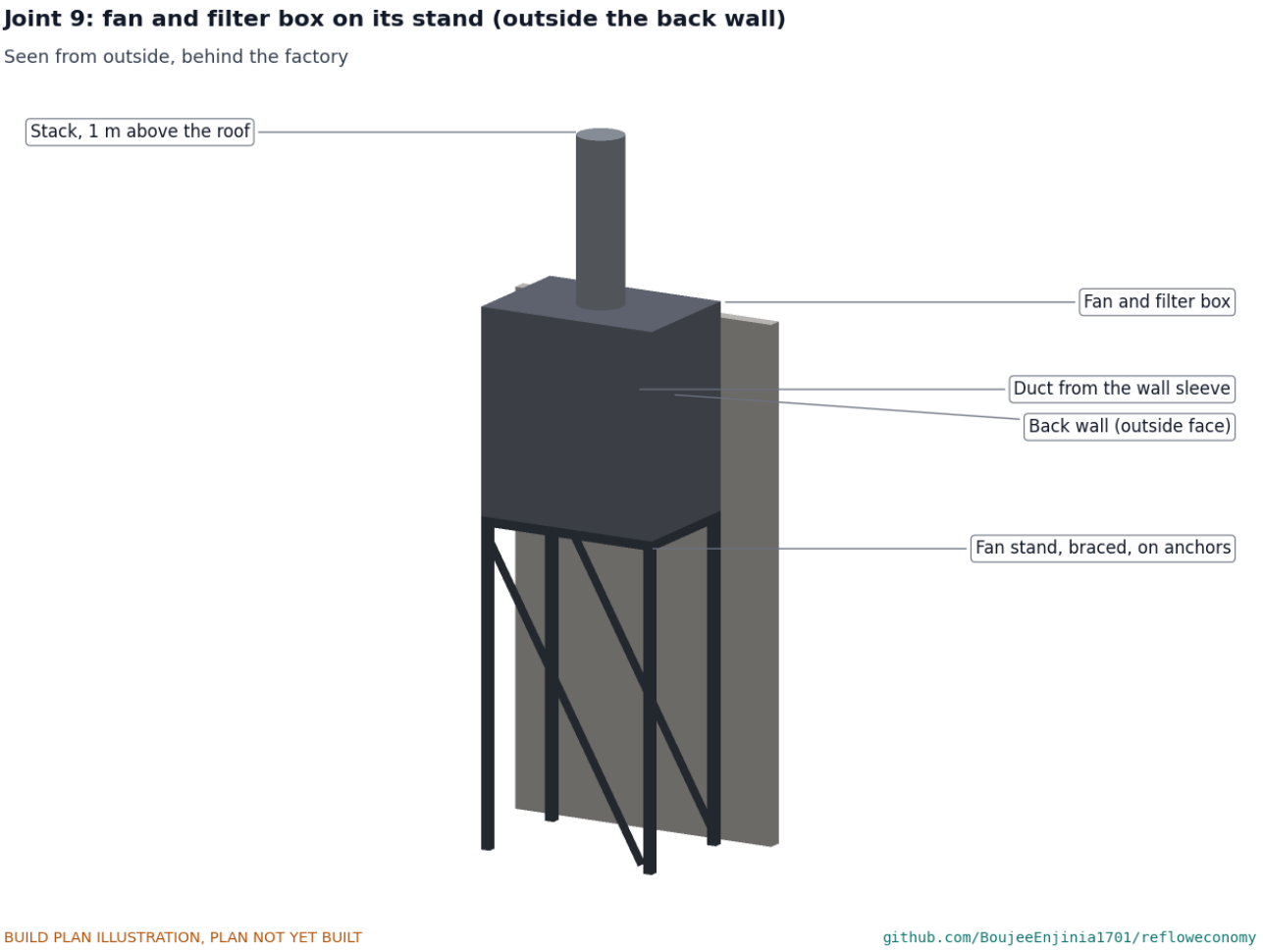


Figure 25. The stand is bolted to a pad; the fan box bolts to its top frame; the stack goes on the outlet.

The stand stands on a concrete pad (or four paving slabs on compacted ground) outside the back wall, its near legs 80 from the wall, centred on the duct hole, anchored with M10 anchors. The fan box bolts to the top frame with four M10 bolts; its inlet takes the duct from the wall sleeve at 2,050 above the inside floor.

Check before moving on. The frame is level and does not rock when pushed by hand at the top.

3.15 Shredder (built to its published design)

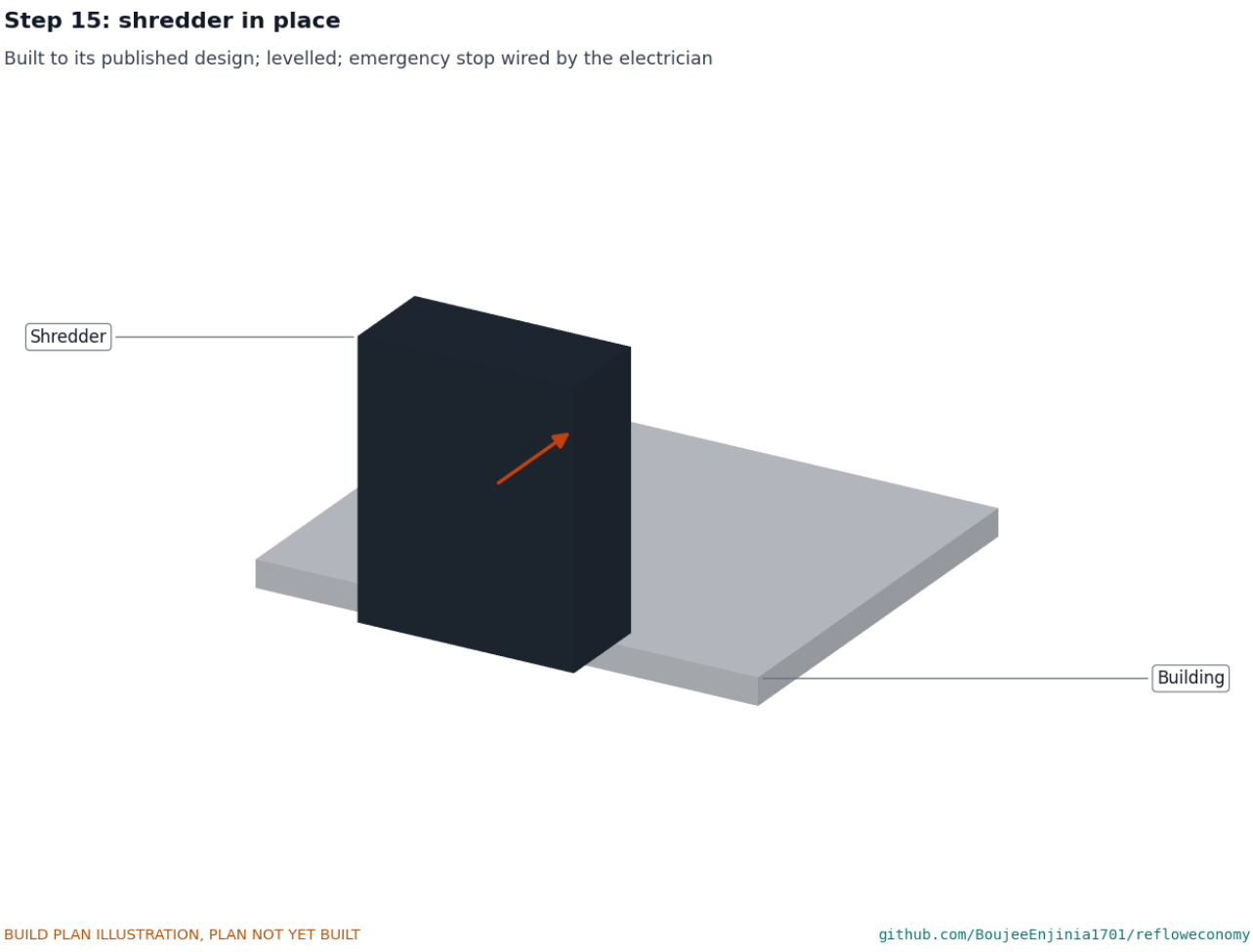


Figure 26. The shredder stands in the front row before its enclosure is built round it.

What to buy or have built. An open-design shredder of the Shredder Pro class, about 1,205 x 550 x 1,512 high including its hopper, with a 2.2 kW gearmotor on a single-phase input drive and an emergency stop. It stands at 6,000 to 7,205 along, 580 to 1,130 from the front wall, levelled on its feet.

3.16 Shredder acoustic enclosure, door and feed chute

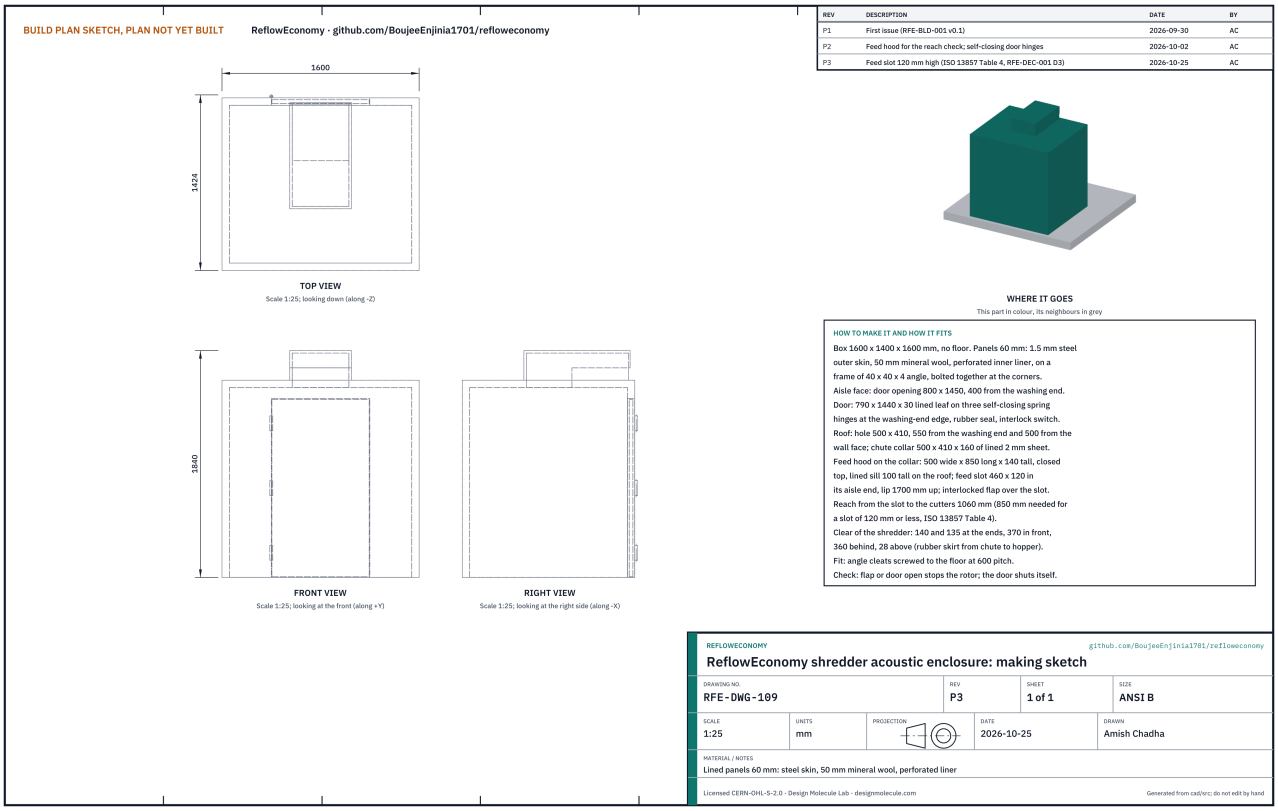


Figure 27. Shredder acoustic enclosure making sketch (RFE-DWG-109).

What it is and what it is made from. A lined box built round the shredder to cut its noise by about 15 dB, with a self-closing interlocked access door and an interlocked feed chute. A feed hood on top of the chute turns the way in, so a hand pushed into the feed slot cannot reach the cutters. Panels 60 thick: 1.5 mm steel outer skin, 50 mm mineral wool and a perforated inner liner, on frames of 40 x 40 x 4 angle.

How to make it.

1. Make four wall panels (two 1,600 long and two 1,400 long, all 1,540 tall) and a roof panel 1,600 x 1,400. There is no floor.
2. In the aisle-side panel, frame a door opening 800 wide x 1,450 tall, 400 from the washing end.
3. Make the door leaf 790 x 1,440 x 30 the same way, with three self-closing spring hinges on the edge nearer the washing tanks, a rubber seal and an interlock switch.
4. In the roof, frame a hole 500 x 410, 550 from the washing end and 500 from the face nearest the front wall.
5. Make the chute collar: a 500 x 410 x 160 tube of lined 2 mm sheet.
6. Make the feed hood: a lined 2 mm sheet box 500 wide, 850 long and 140 tall with a closed top, closed sides and a closed end over the collar, and a lined sill 100 tall under its other half that stands on the roof. Its open end, facing the aisle, is the feed slot, 460 wide and 120 tall, with its bottom lip 1,700 above the floor. Keep the slot no taller than 120: above that, the safety distance in ISO 13857 for reaching through an opening no longer applies. Hang a flap over the slot from the top edge, with an interlock switch.

How it fits the parts next to it.

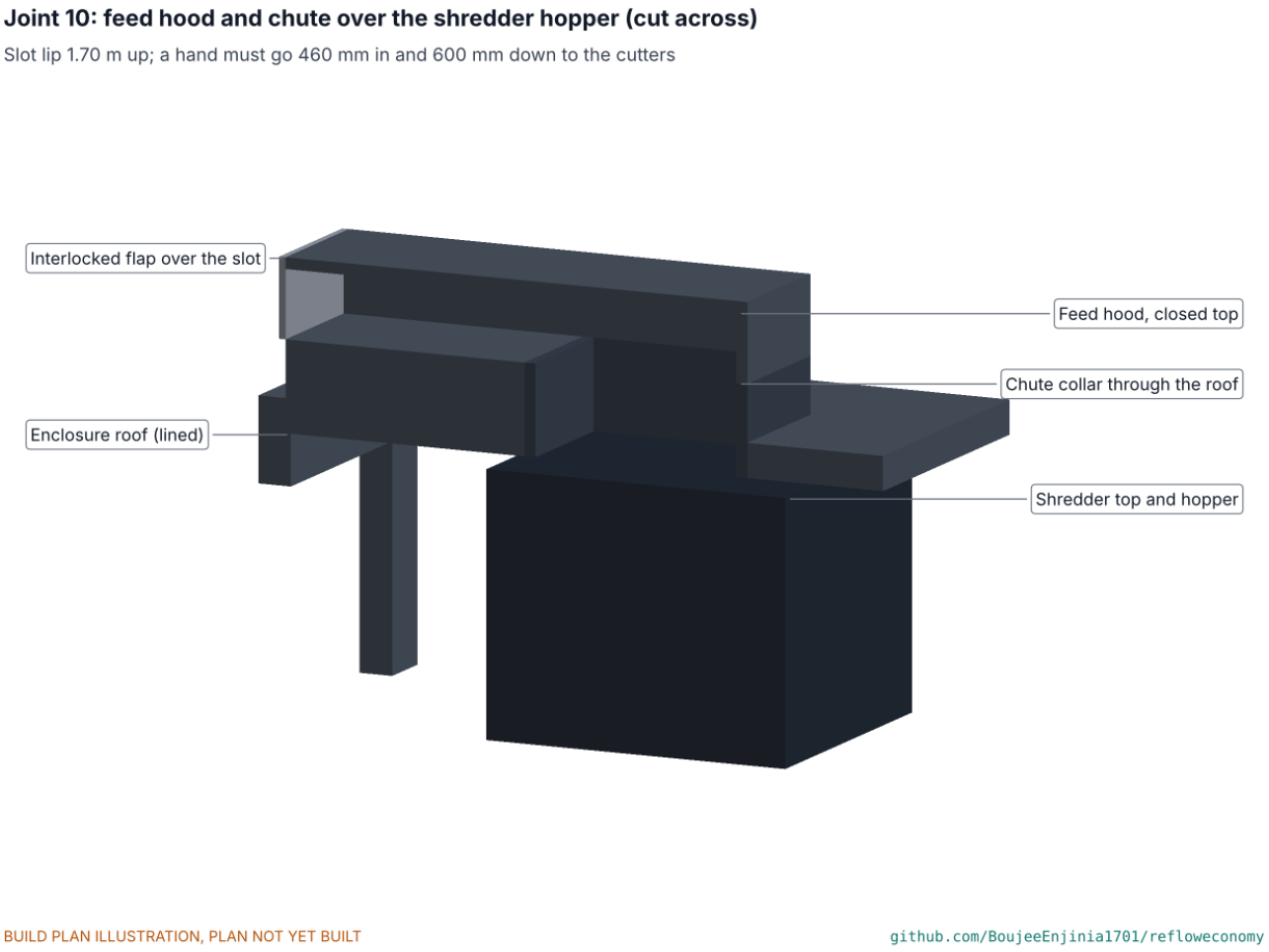


Figure 28. Cut across: material goes in through the slot, along the hood and down the collar to the hopper. A hand would have to go 460 mm in and 600 mm down, 1,060 mm in all, to touch the cutters; 850 mm is enough for a slot 120 mm tall or less. A rubber skirt closes the 28 mm gap below the collar.

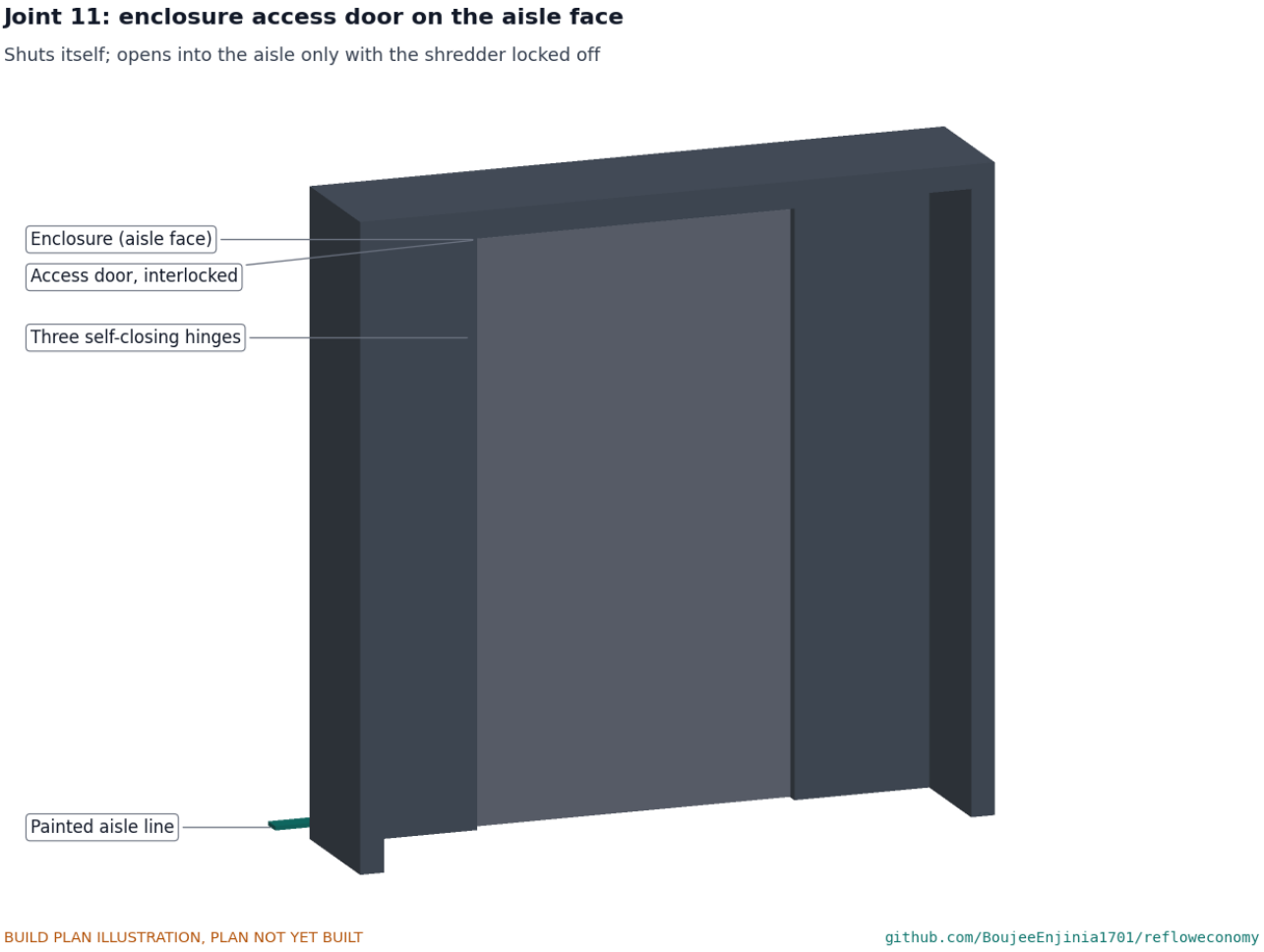


Figure 29. The door shuts itself on its spring hinges and opens into the aisle only with the shredder locked off; the flake bin comes out here.

The panels bolt together at the corners round the shredder and stand on the floor at 5,800 to 7,400 along and 150 to 1,550 from the front wall, held by angle cleats screwed down at 600 pitch. They clear the shredder by 140 and 135 at the ends, 370 in front, 360 behind and 28 above. The collar sits in the roof hole and the feed hood bolts on top of it, its sill on the roof, with the slot facing the aisle and its lip 1,700 above the floor. Feed pieces must be thinner than the 120 mm slot: crush bottles flat and cut bigger parts at the sorting table. Both interlock switches are wired into the shredder's stop circuit by the electrician. The reach from the slot to the cutters assumes the cutters are about 1,100 above the floor; measure them on the machine actually built and tell the safety professional if they are higher.

Check before moving on. Opening the flap, or the door, stops the rotor every time, and the door swings shut by itself when let go (stop S5).

3.17 Washing tanks and trap (bought)

Two 400 to 500 L polyethylene tanks (pre-wash and float-sink rinse) on level floor at 3,800 to 4,600 and 4,700 to 5,500 along, 300 to 1,000 from the front wall; a settling trap and a splash-rated 0.75 kW pump on a skid in front of them, 1,100 to 1,600 from the front wall. The pump plugs into a residual-current-protected socket fed from the front cable tray, above splash height. See step 17.

3.18 Drying rack and fan stand

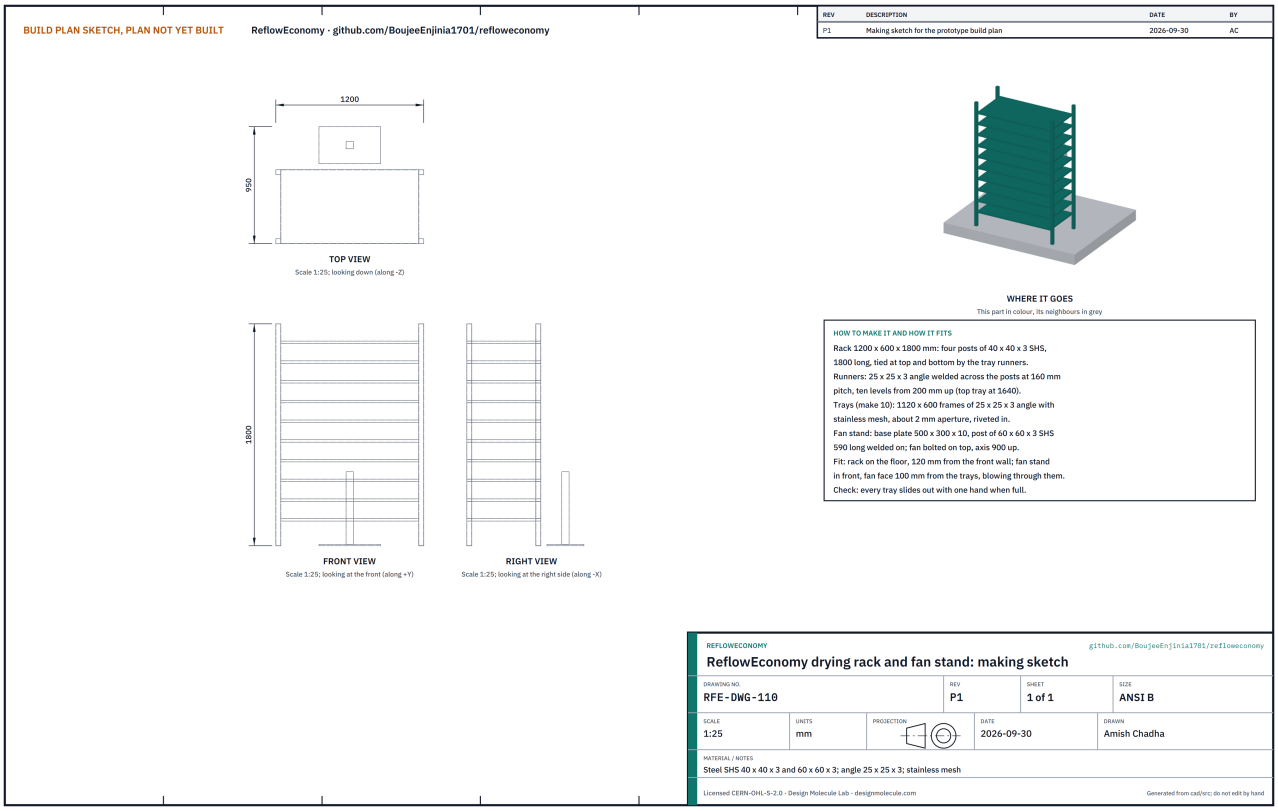


Figure 30. Drying rack and fan stand making sketch (RFE-DWG-110).

What it is and what it is made from. A rack of ten mesh trays (7.2 m² in all) for washed flake, and a stand for the 0.4 kW fan that blows through it. Steel square tube 40 x 40 x 3 and 60 x 60 x 3, angle 25 x 25 x 3, stainless mesh of about 2 mm aperture.

How to make it.

1. Cut four posts of 40 x 40 x 3 tube, 1,800 long, and set them out on a 1,200 x 600 rectangle.
2. Weld runners of 25 x 25 x 3 angle across the short sides at 160 pitch, ten levels from 200 up (the top tray sits at 1,640).
3. Make ten trays, each a 1,120 x 600 frame of 25 x 25 x 3 angle with mesh riveted in.
4. Fan stand: weld a 590 long post of 60 x 60 x 3 tube to a 500 x 300 x 10 base plate, with a plate on top to take the fan.

How it fits the parts next to it.

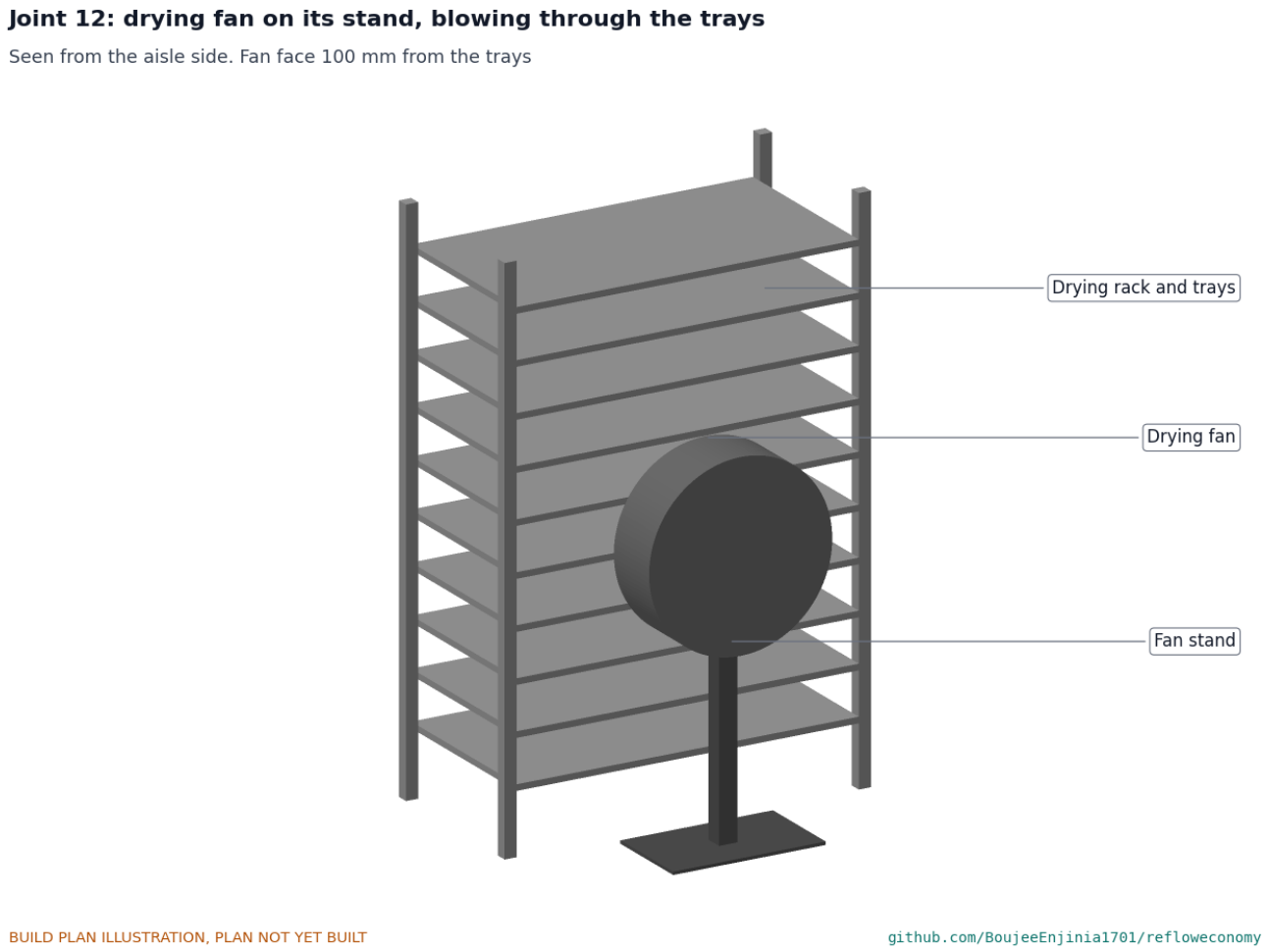


Figure 31. Seen from the aisle side: the fan face is 100 mm from the trays.

The rack stands on the floor at 7,700 to 8,900 along, 200 to 800 from the front wall. The fan bolts to the stand, axis 900 above the floor, blowing through the trays toward the front wall.

Check before moving on. Every full tray slides out with one hand.

3.19 Intake, racking and passport desk (bought)

- **Intake.** A 300 kg platform scale 1 x 1 m at 300 to 1,300 along; a steel sorting table 2.0 x 0.9 m, 900 high, from 110 to 1,010 from the front wall; four colour-coded bins 420 x 600 x 800 (PET, HDPE, PP, metals) behind it, 1,030 to 1,630 from the front wall, behind the aisle line (step 18).
- **Racking.** Two pallet racking bays 1.35 x 1.0 x 2.0 m with three levels, anchored to the floor: bay 1 (products) in the back row at 10,750 to 12,100 along, bay 2 (flake bags) in the front row at 9,100 to 10,450 along. Bay 1 is exactly 1.0 m from the hot zone line.
- **Passport desk.** A desk 1,100 x 700 at 10,800 to 11,900 along with a thermal label printer, a 30 kg bench scale, a moisture meter and a laptop or phone.

3.20 Export and residue cage

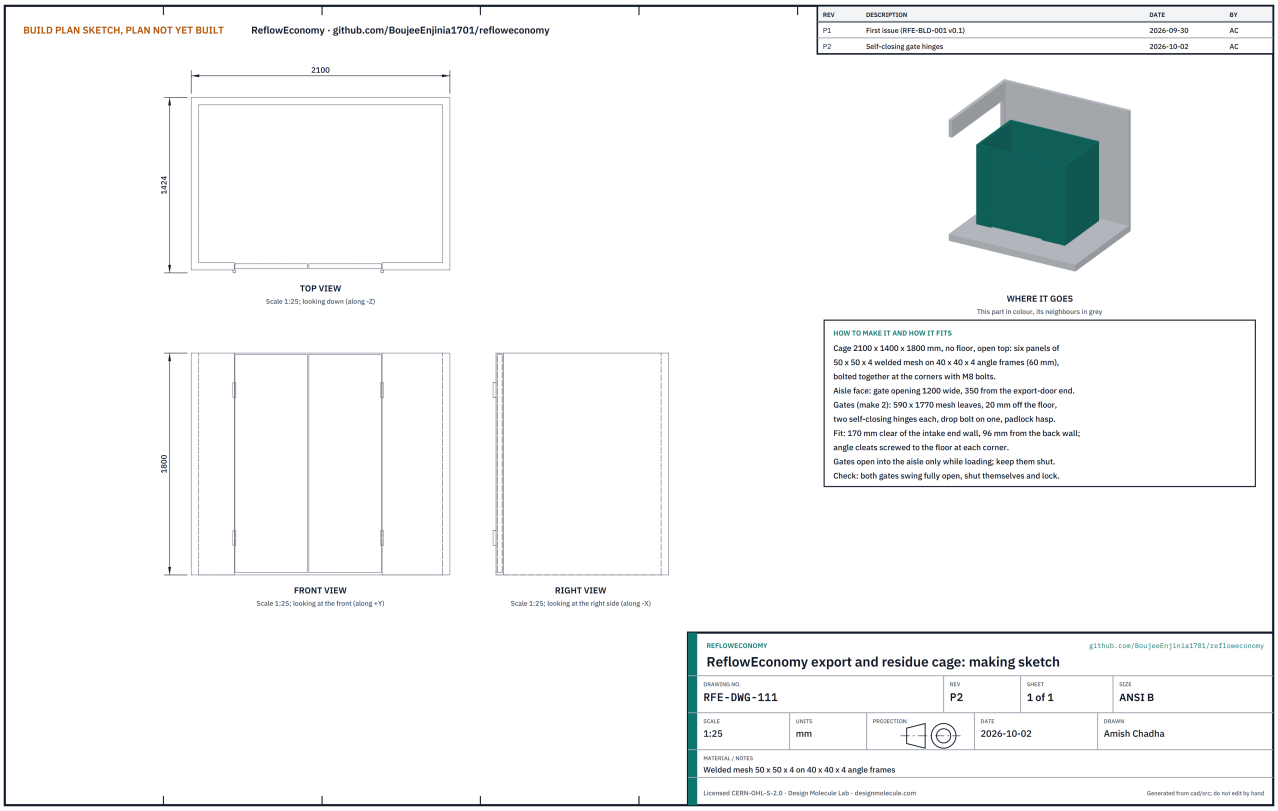


Figure 32. Export and residue cage making sketch (RFE-DWG-111).

What it is and what it is made from. A lockable cage by the export doors for metals, boards, cells and residue waiting to leave. Welded mesh 50 x 50 x 4 on frames of 40 x 40 x 4 angle.

How to make it.

1. Make six panels 1,800 tall: two sides 1,400 wide, a back 2,100 wide, and on the aisle face two panels either side of a 1,200 gate opening (350 and 550 wide).
2. Make two gate leaves 590 x 1,770 of the same mesh and angle, each on two self-closing gate hinges, with a drop bolt on one and a padlock hasp on the other.
3. Drill the frames for M8 bolts at the corners.

How it fits the parts next to it.

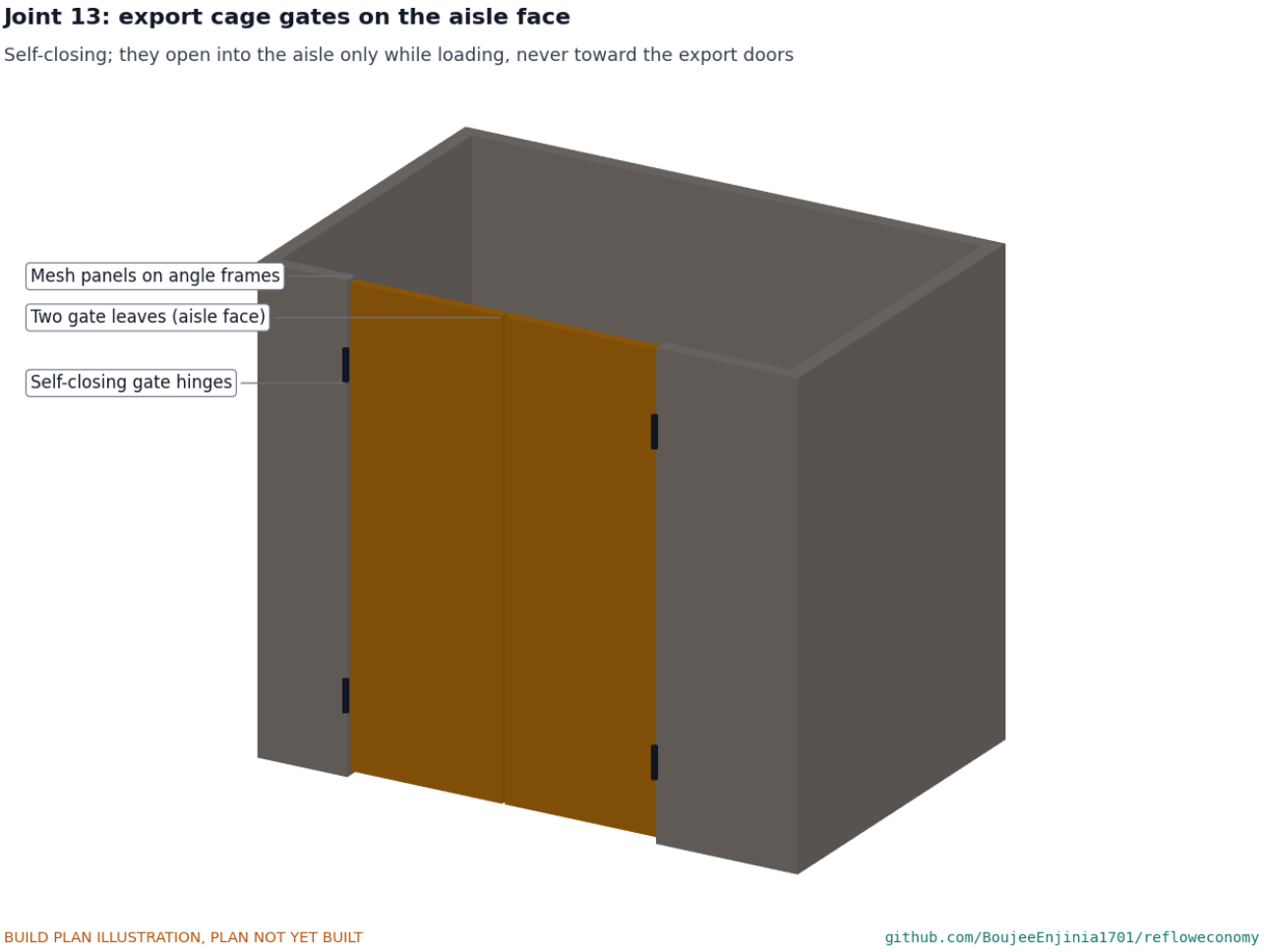


Figure 33. The gates shut themselves and open into the aisle only while loading, not toward the export doors 170 mm away.

The cage stands at 250 to 2,350 along and 3,300 to 4,700 from the front wall, no floor, open top, held by angle cleats at each corner. Inside go a fire-safe steel container with dry sand for lithium cells and the bale bags.

Check before moving on. Both gates swing fully open, shut by themselves when let go, and lock shut. With them open, 1.08 m of aisle remains.

3.21 Bought components

Buy to specification, not brand. Item numbers are those of the bill of materials.

- **Building shell (item 13).** A rented shed of about 60 m² with a concrete floor, or two used 40 ft containers with the seam work of section 3.1.
- **Intake (item 1), washing (item 2).** As sections 3.19 and 3.17.
- **Shredder (item 3).** As section 3.15.
- **Extruder (item 5), presses (item 6).** As sections 3.7 and 3.9; HDPE and PP only.
- **Fume extraction (item 7).** Duct, fittings, fan and filter box and stack as sections 3.13 and 3.14.

- **Racking (item 8), passport desk (item 9), safety station (item 10).** As sections 3.19 and 3.4; PPE for four workers (respirators, face shields, heat and puncture-resistant gloves, hearing protection, safety boots), first aid kit, sharps container.
- **Electrical board and tray (item 11).** As section 3.6, installed by a licensed electrician.
- **Fixings.** M8 bolts, nuts and washers; M8 and M10 rivet nuts; M10 concrete anchors; M6 countersunk wood screws; steel rivets; closed-cell foam strip; duct sealant and tape; rubber duct collar.

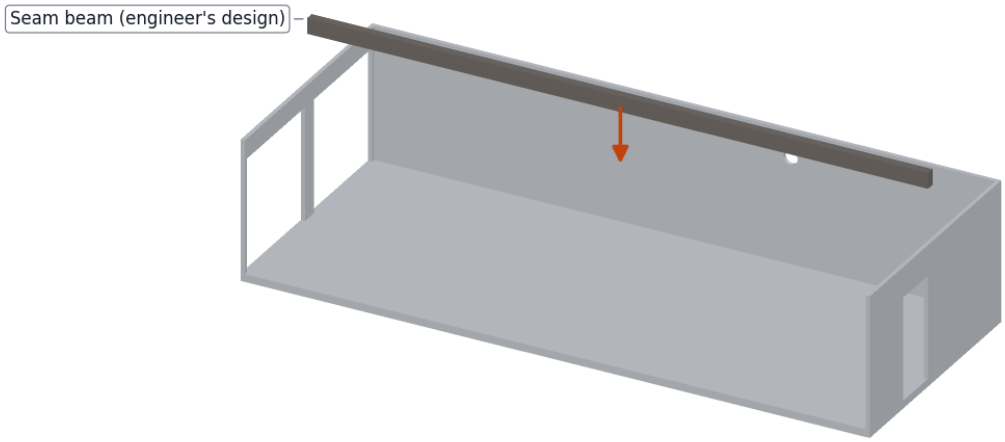
4. Putting it together

In each picture the parts already fitted are grey and the parts being fitted are in colour, with an arrow showing the way they go in.

Step 1: seam beam in, then the side walls out (container option)

Step 1: seam beam in, then the side walls out (container option)

Props and beam to the engineer's design and sequence; only then are the joining side walls cut out



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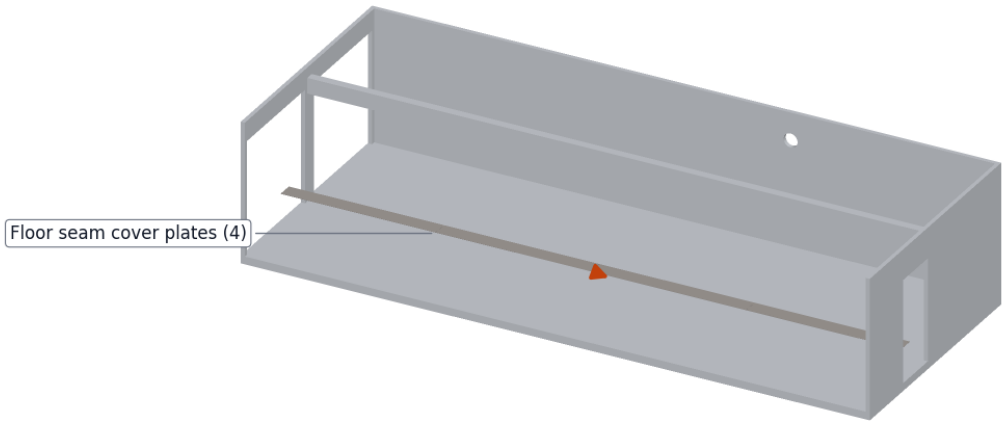
github.com/BoujeeEnjinia1701/refloweconomy

Containers on level footings, doors at the intake end. The fabricator props the roofs, fits the seam beam to the engineer's design, and only then cuts out the joining side walls and flashes the roof joint. **Hold point:** stop S1.

Step 2: floor seam cover plate

Step 2: floor seam cover plate

Screwed along one edge only, into the intake-side container floor



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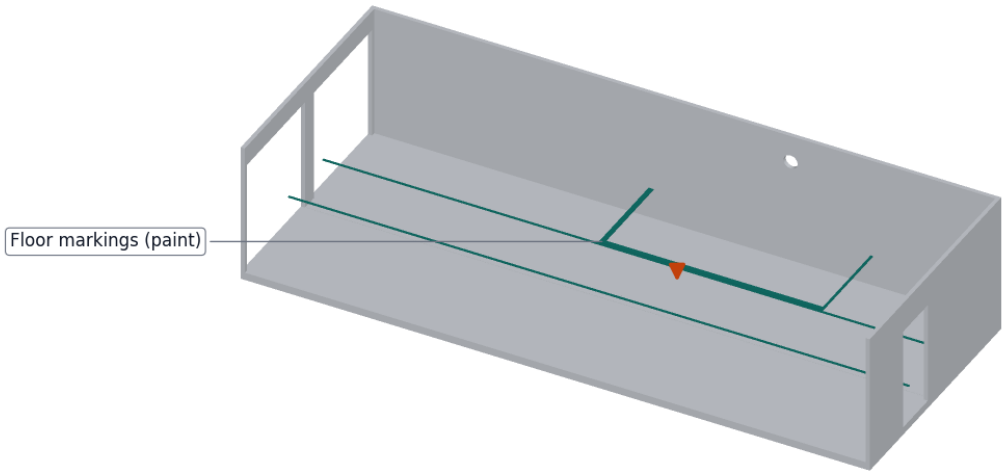
github.com/BoujeeEnjinia1701/refloweconomy

Four lengths, drilled edge over the intake-side container, M6 countersunk screws into that floor only.

Step 3: paint the aisle lines and the hot zone boundary

Step 3: paint the aisle lines and the hot zone boundary

50 mm lines; the hot zone line is 1.0 m from any stock



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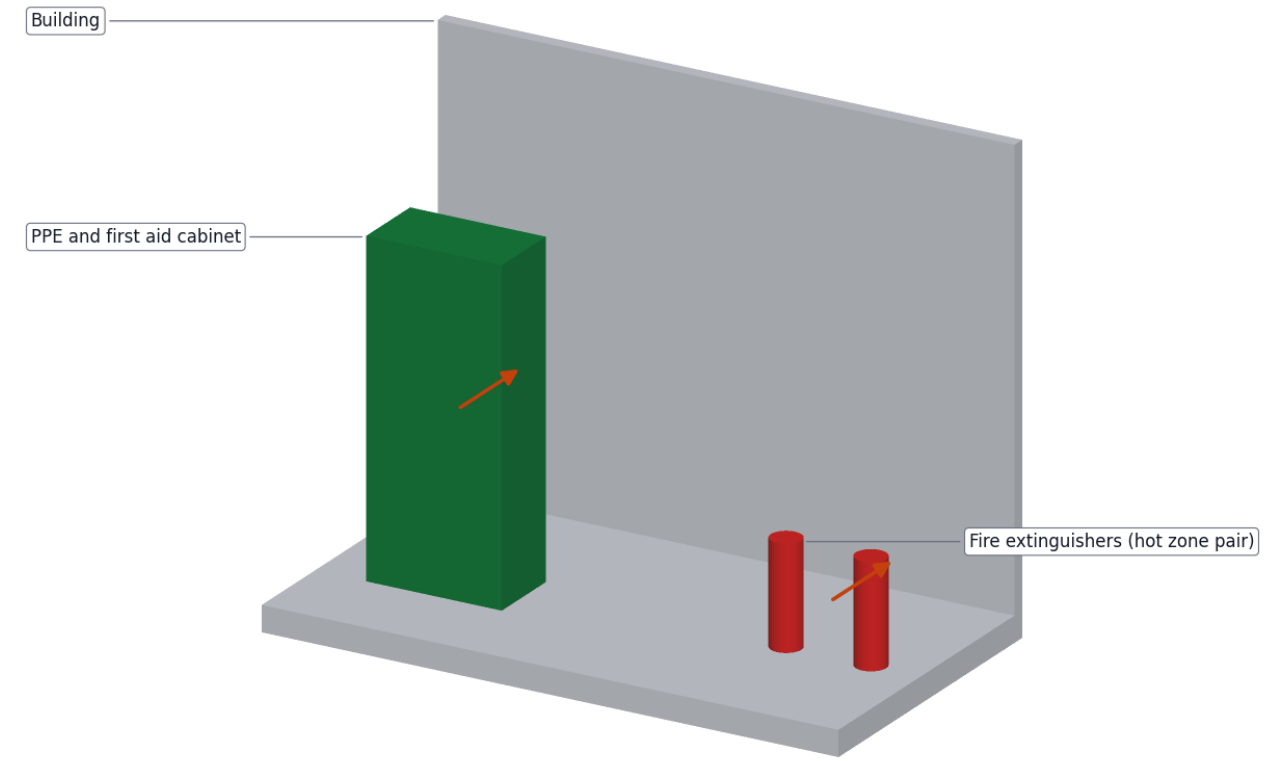
github.com/BoujeeEnjinia1701/refloweconomy

Mark out from the front wall and the intake end wall as section 3.3; two coats.

Step 4: safety station first

Step 4: safety station first

Cabinet against the back wall on two brackets; extinguishers in place before any power work



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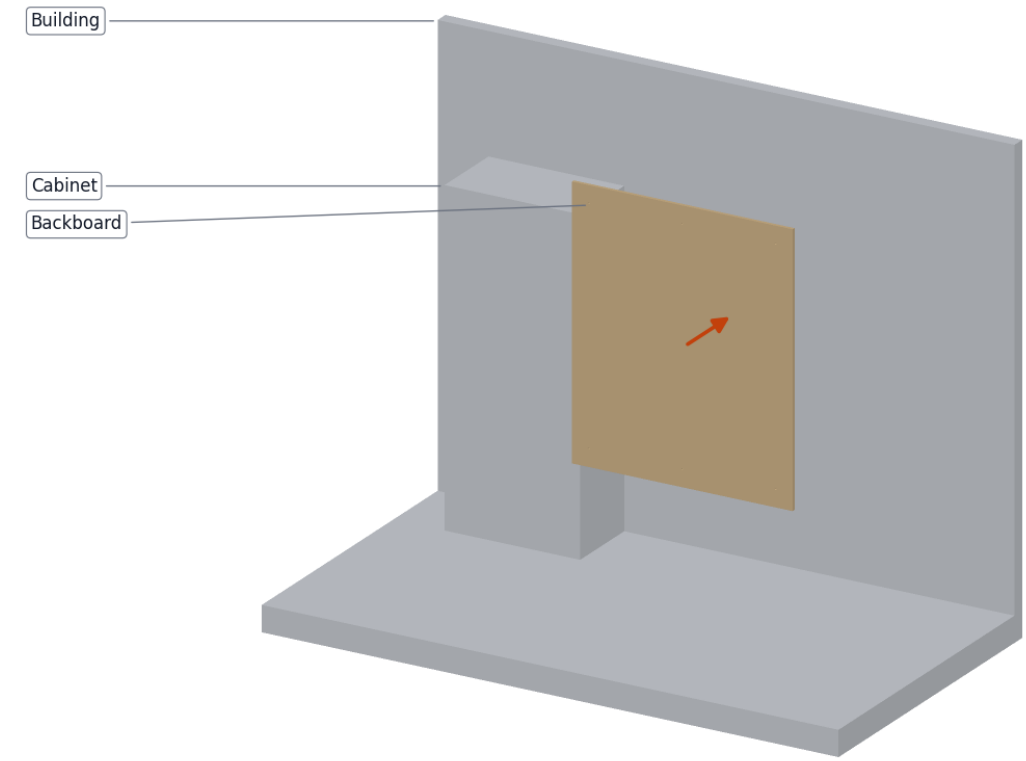
github.com/BoujeeEnjinia1701/refloweconomy

Cabinet against the back wall on two brackets; the two hot zone extinguishers on their stands. These are in place before any power or hot work.

Step 5: backboard onto the back wall

Step 5: backboard onto the back wall

Six M8 screws into rivet nuts in the corrugation crests



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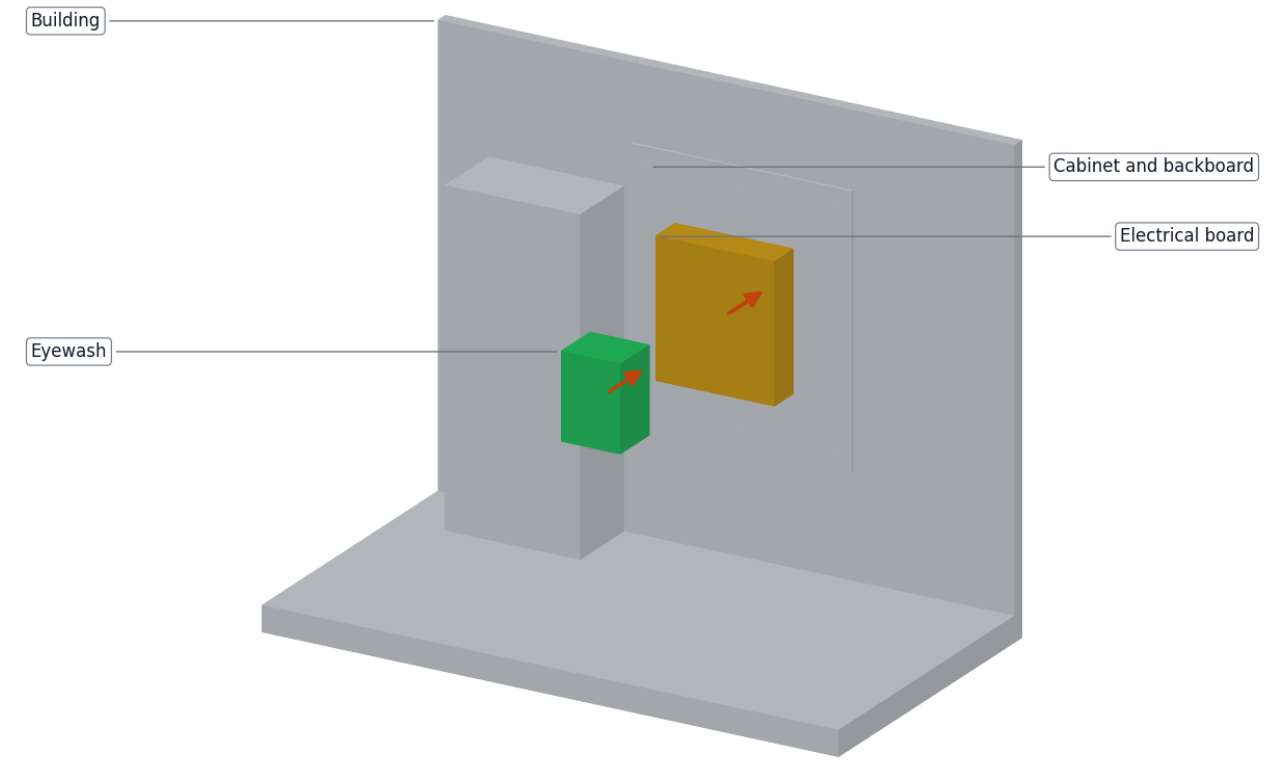
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Six M8 screws into rivet nuts in the corrugation crests.

Step 6: eyewash and electrical board onto the backboard

Step 6: eyewash and electrical board onto the backboard

The licensed electrician fits the board, RCDs, emergency stop circuit and heater interlock



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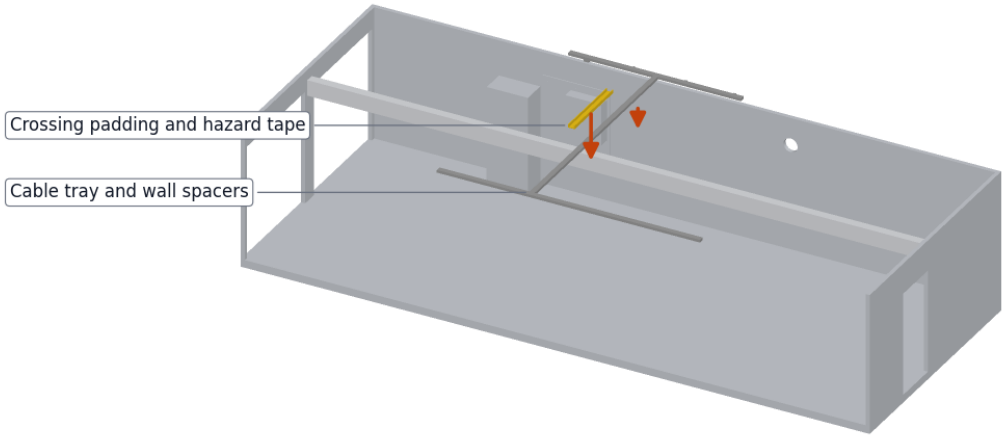
github.com/BoujeeEnjinia1701/refloweconomy

The electrician fits the board with its residual current devices, emergency stop circuit and heater interlock; the eyewash bracket screws to the backboard beside it.

Step 7: cable tray

Step 7: cable tray

Back run, crossing under the seam beam with taped padding over the aisle, front run



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

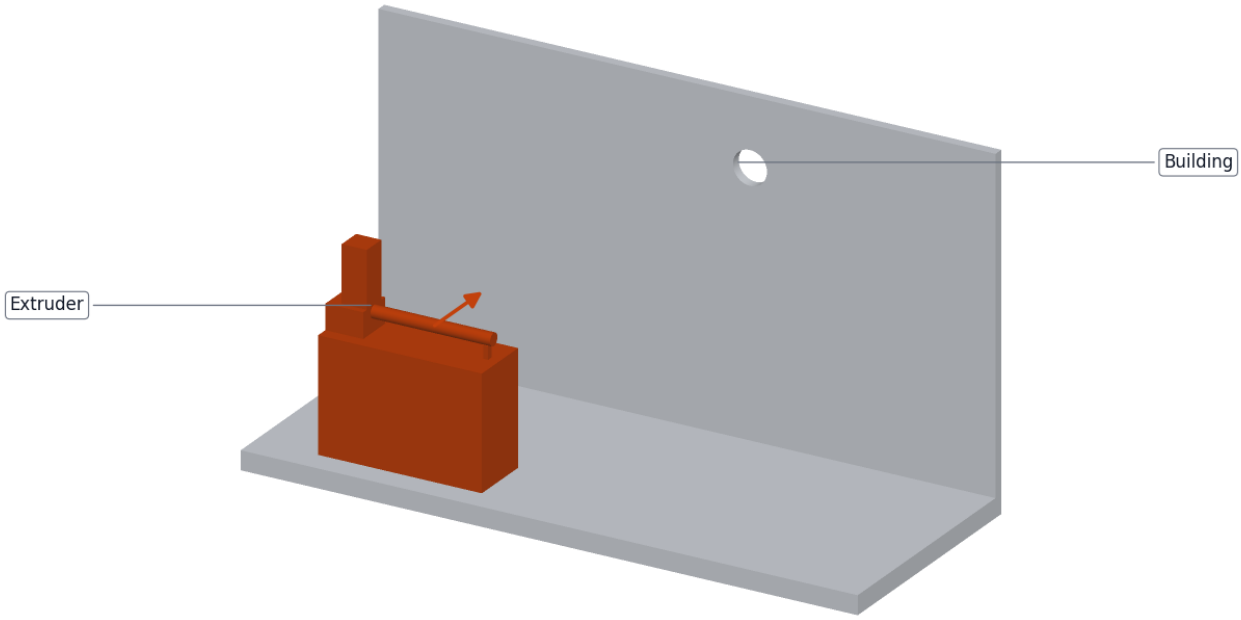
github.com/BoujeeEnjinia1701/refloweconomy

Spacers, then the back run, the crossing under the seam beam on two beam clamps, and the front run. Then the foam padding and hazard tape on the crossing over the aisle. **Hold point:** stop S2.

Step 8: extruder into the hot zone

Step 8: extruder into the hot zone

Built to its published design by the fabricator; levelled on its feet



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

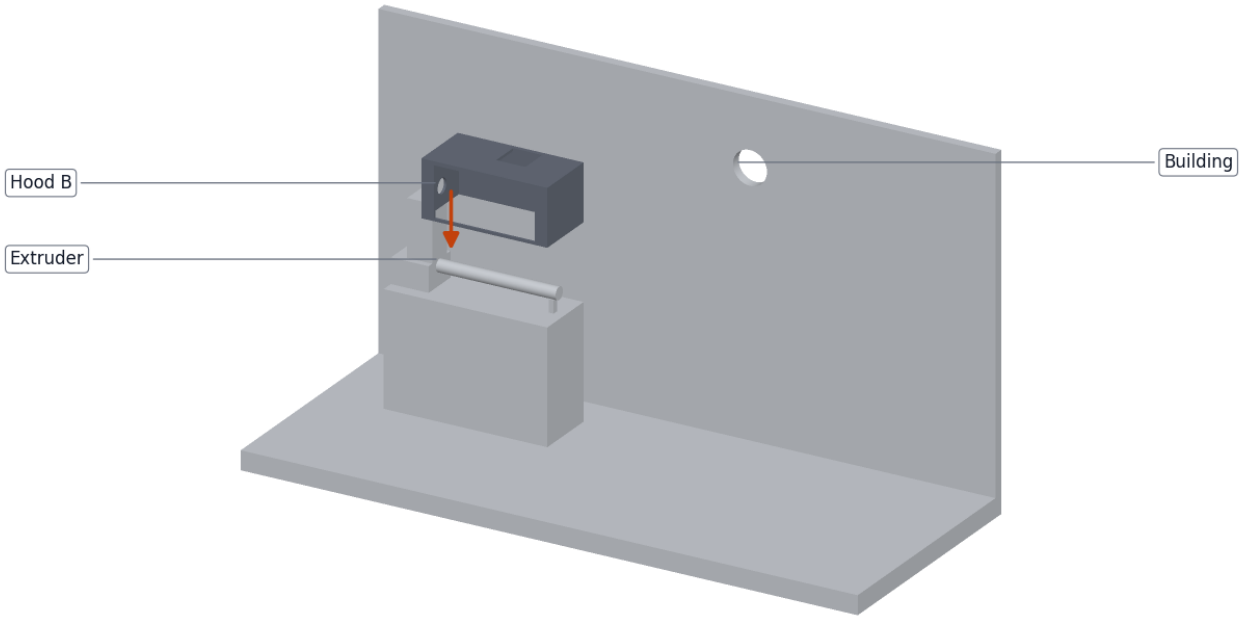
github.com/BoujeeEnjinia1701/refloweconomy

Levelled on its feet, barrel along the line, hopper toward the intake end.

Step 9: hood B onto the extruder frame

Step 9: hood B onto the extruder frame

Six M8 bolts through its bottom flange; the barrel passes through the end hole



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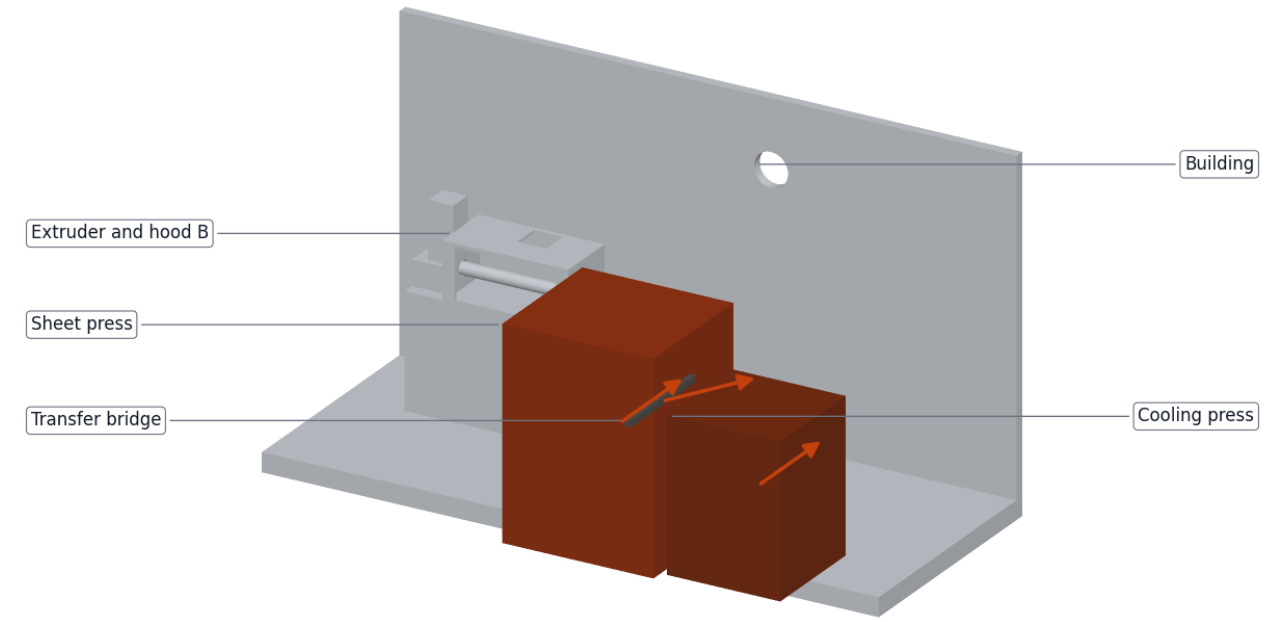
github.com/BoujeeEnjinia1701/refloweconomy

Lowered over the barrel so the barrel passes through the end hole; six M8 bolts through the bottom flange.

Step 10: presses and transfer bridge

Step 10: presses and transfer bridge

Both presses in before the booth; bridge bolted to the hot press, roller level with the platens



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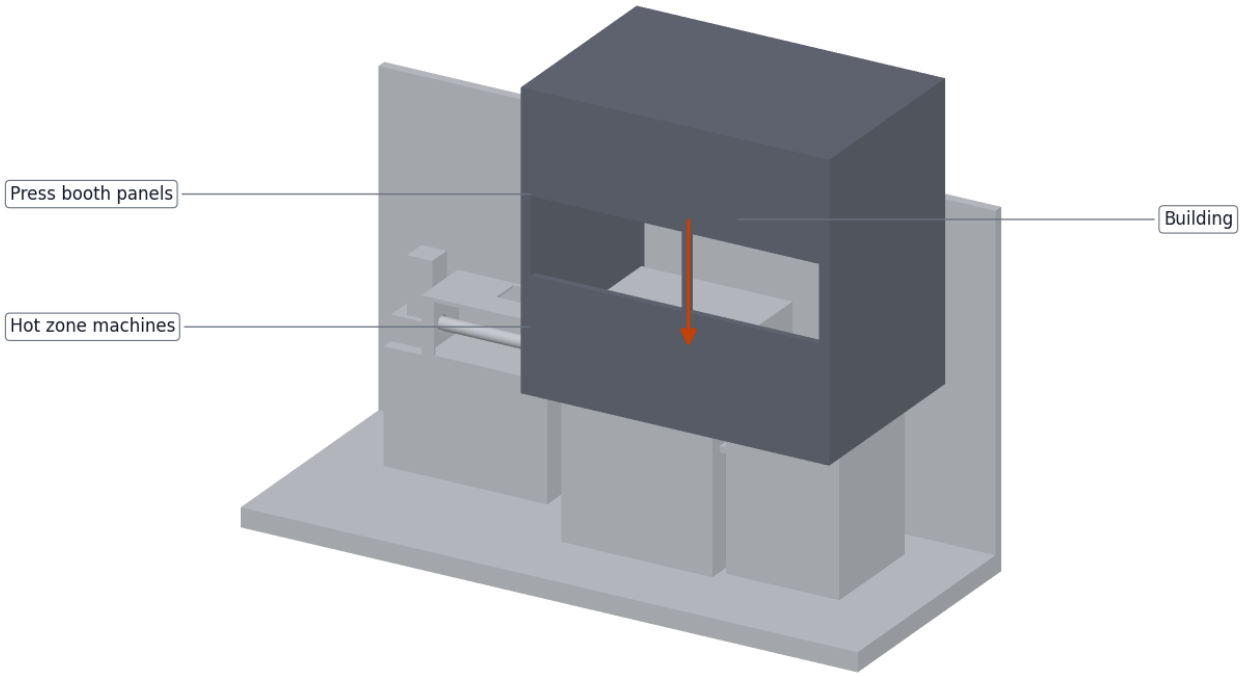
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Both presses in place and levelled before the booth goes up; the bridge bolted to the hot press frame, its roller level with both platens.

Step 11: press booth round the presses

Step 11: press booth round the presses

Panels bolted together, cleated to the floor, back edges sealed to the wall



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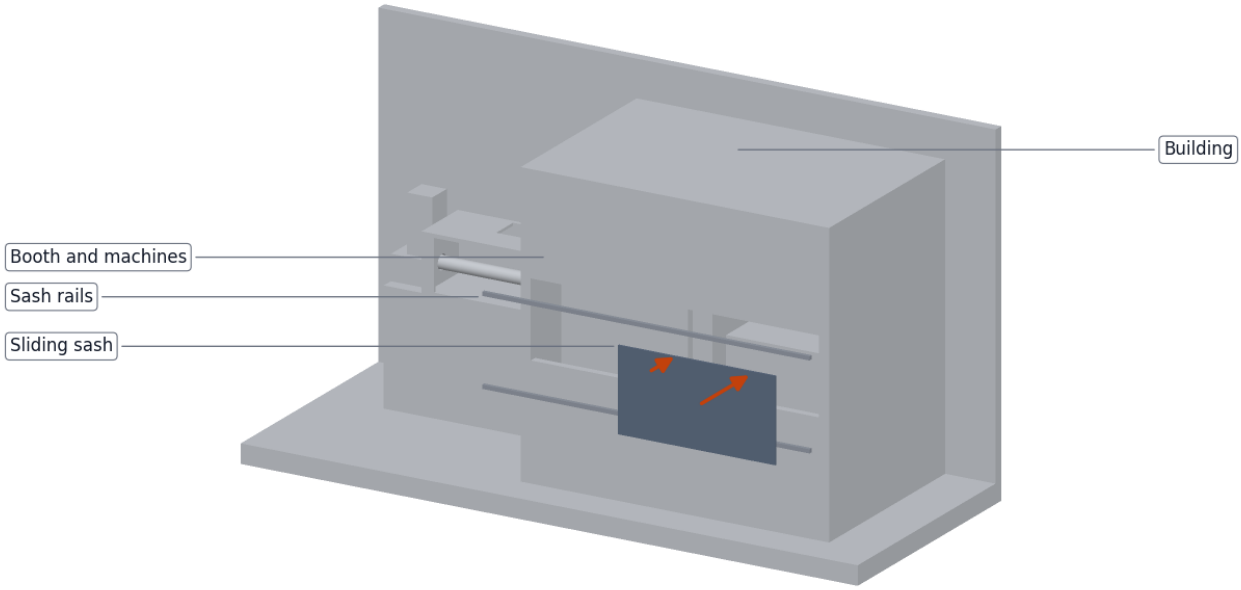
github.com/BoujeeEnjinia1701/refloweconomy

Ends first, then the front, then the roof, all bolted through their frames; cleats screwed to the floor; back edges on foam against the wall.

Step 12: sash rails and the sliding sash

Step 12: sash rails and the sliding sash

Rails screwed to the booth front; the sash is slid in from the right-hand end



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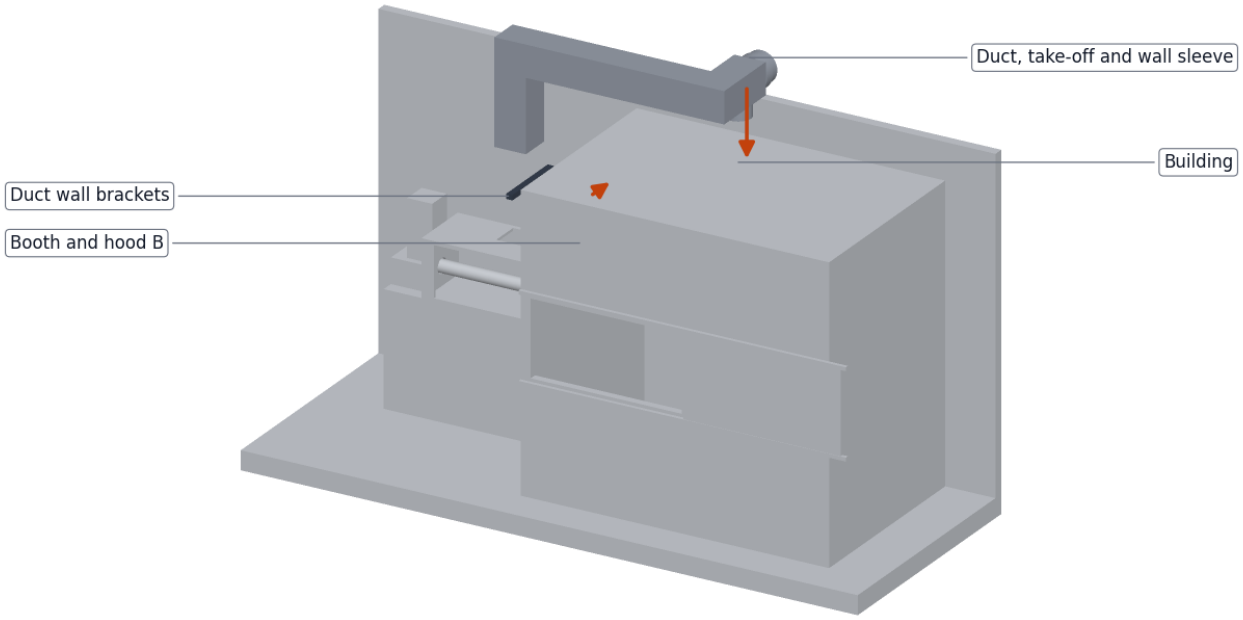
github.com/BoujeeEnjinia1701/refloweconomy

Rails screwed to the booth front; the sash slid in from the right-hand end; end stops fitted.

Step 13: duct, brackets, booth take-off and wall sleeve

Step 13: duct, brackets, booth take-off and wall sleeve

Brackets first; the duct goes in through the booth end wall collar, then the elbow and sleeve



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

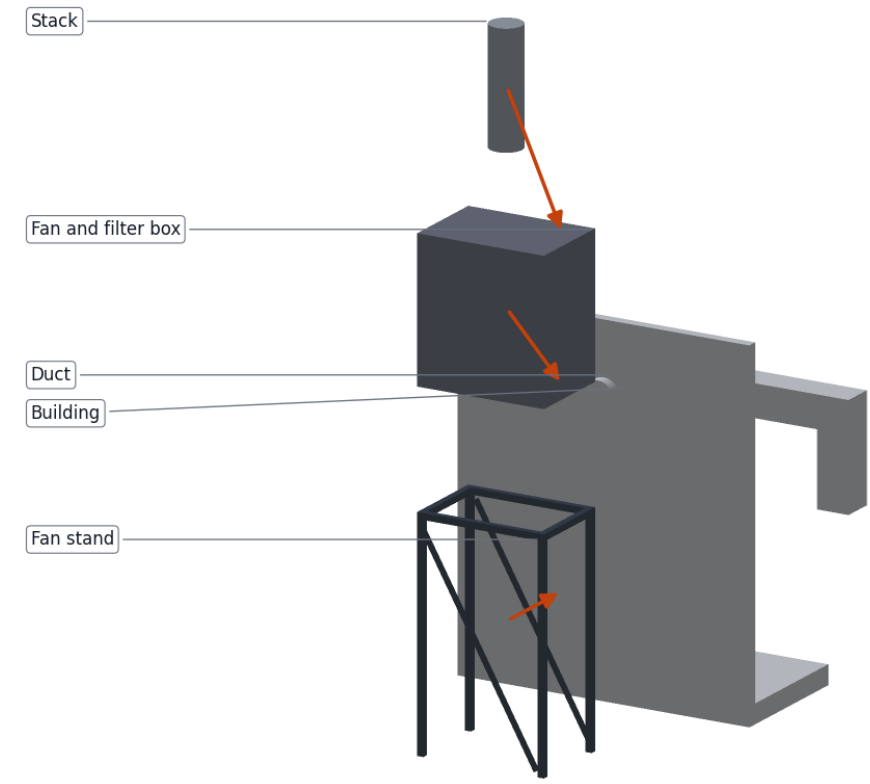
github.com/BoujeeEnjinia1701/refloweconomy

Brackets first; riser onto the hood B spigot; the run through the booth end wall collar onto the brackets; the elbow with its take-off; the sleeve through the back wall. Every joint sealed.

Step 14: fan stand, fan and filter box, stack (outside)

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Seen from behind. Stand anchored to a pad; fan box bolted on; stack 1 m above the roof



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

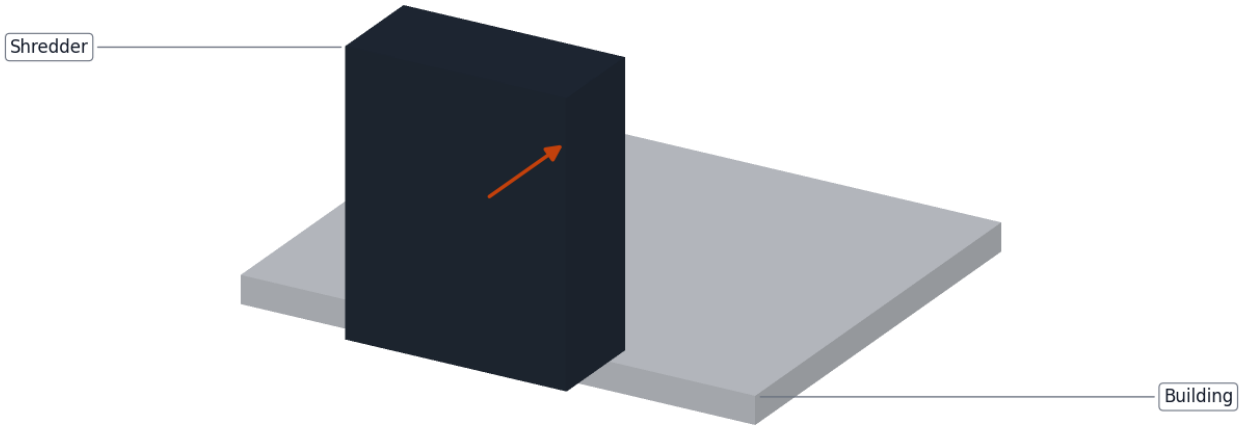
github.com/BoujeeEnjinia1701/refloweconomy

Seen from behind. Stand anchored to its pad; fan box bolted on and joined to the sleeve; stack on the outlet, 1 m above the roof.

Step 15: shredder in place

Step 15: shredder in place

Built to its published design; levelled; emergency stop wired by the electrician

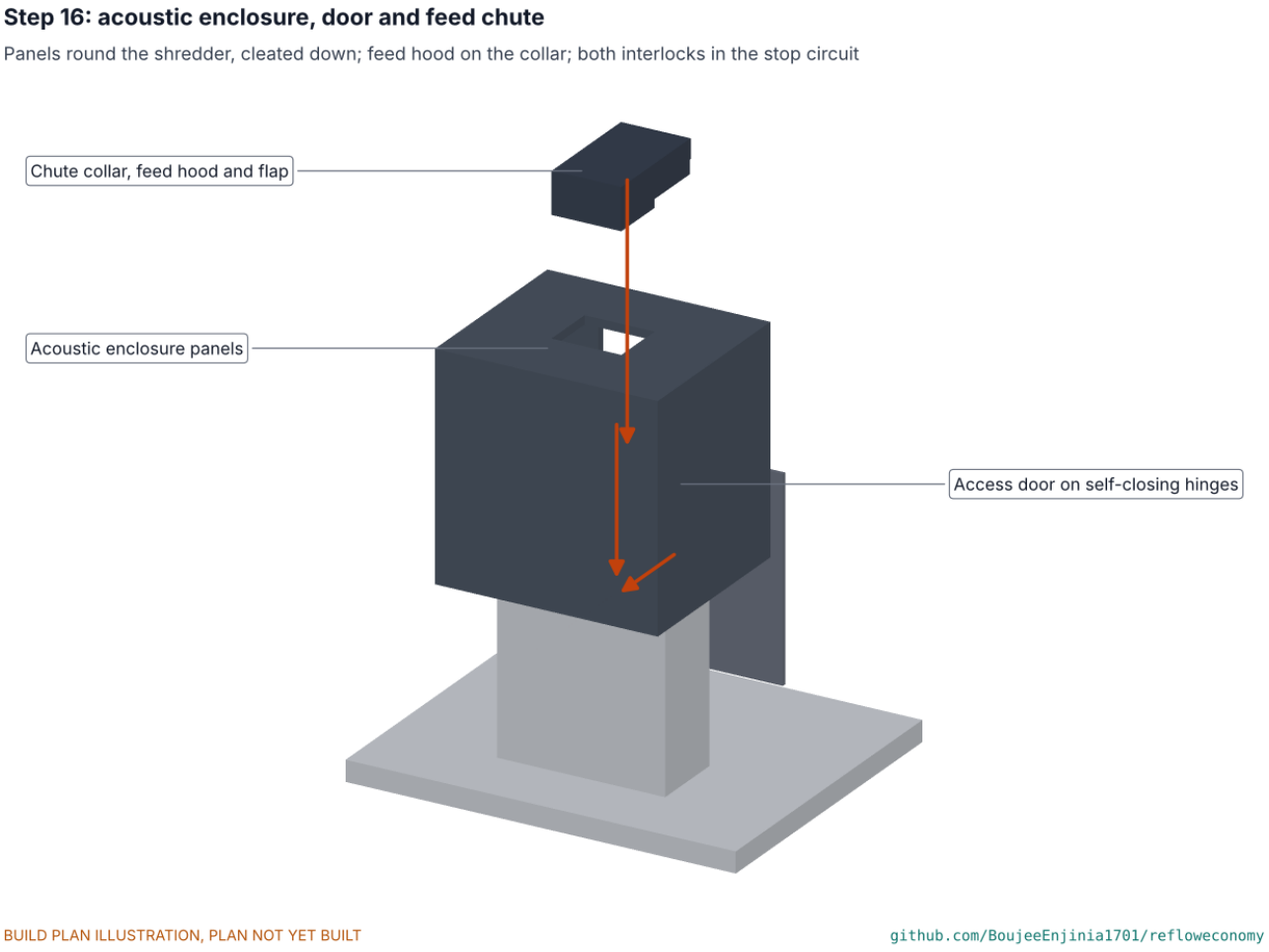


BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/refloweconomy

Levelled; its emergency stop and drive wired from the front tray by the electrician.

Step 16: acoustic enclosure, door and feed chute

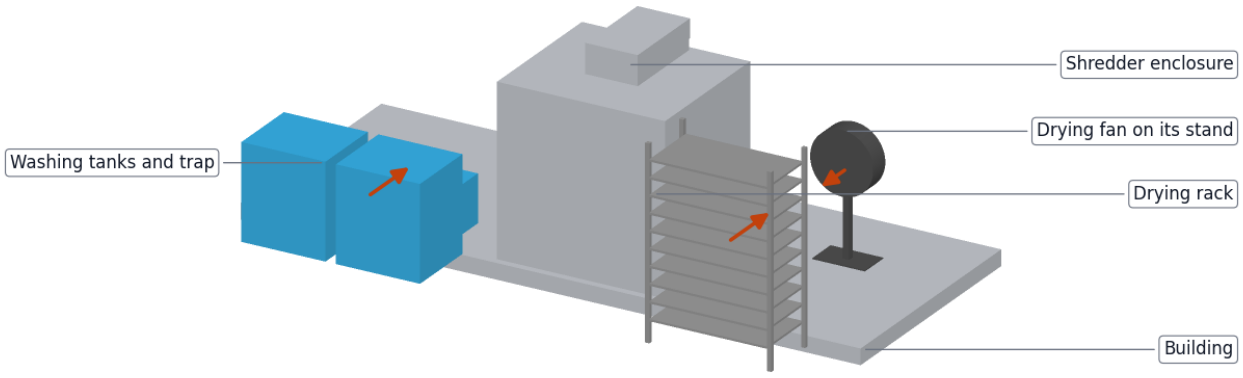


Wall panels bolted together round the shredder and cleated down, then the roof, the collar, the feed hood with its flap, and the door on its self-closing hinges. Both interlocks wired into the stop circuit. **Hold point:** stop S5.

Step 17: washing tanks, drying rack and fan

Step 17: washing tanks, drying rack and fan

Tanks on level ground near the RCD-protected socket; fan stand in front of the rack



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

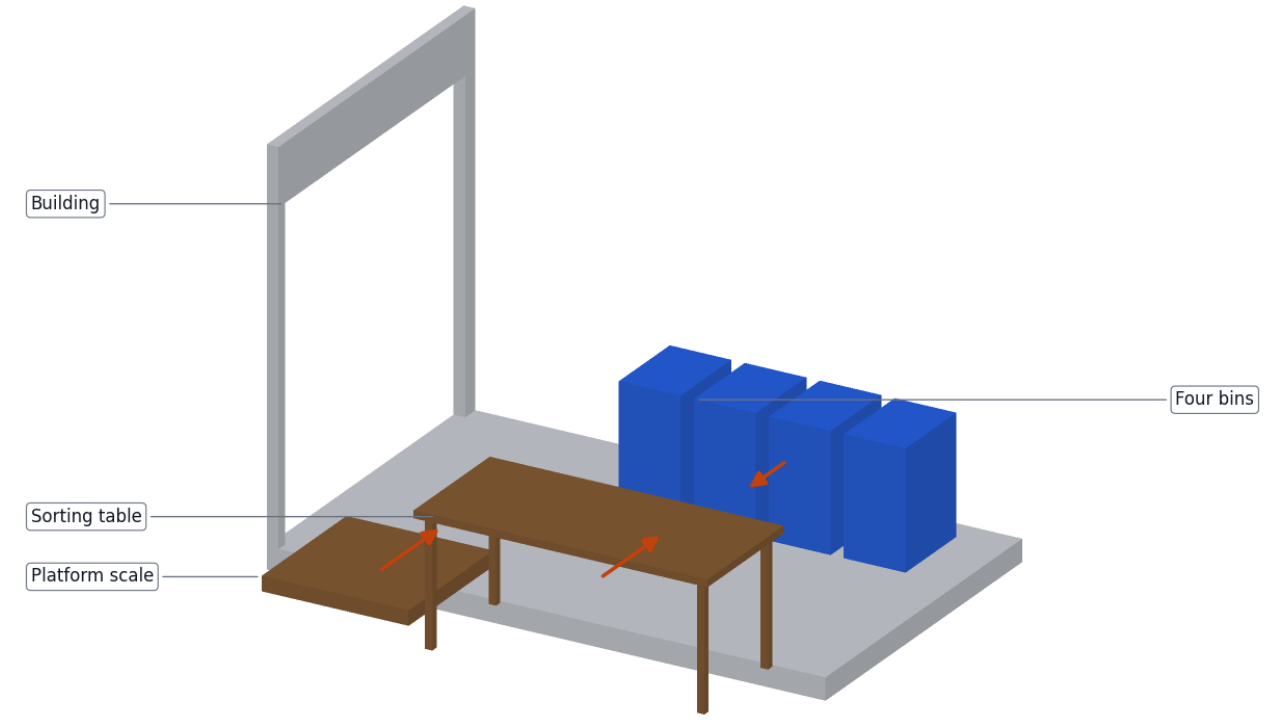
github.com/BoujeeEnjinia1701/refloweconomy

Tanks and pump skid on level floor; rack on the floor 120 from the front wall; fan stand in front of it.

Step 18: intake scale, sorting table and bins

Step 18: intake: scale, sorting table and bins

Bins stand behind the painted aisle line, not on it



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

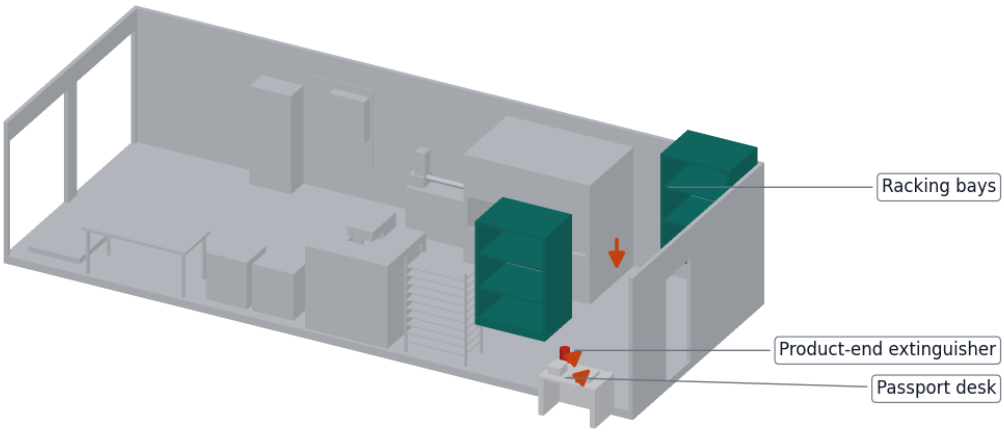
github.com/BoujeeEnjinia1701/refloweconomy

Scale inside the intake doors; table and bins behind the aisle line.

Step 19: racking, passport desk and the last extinguisher

Step 19: racking, passport desk and the last extinguisher

Racking anchored to the floor, 1.0 m clear of the hot zone line

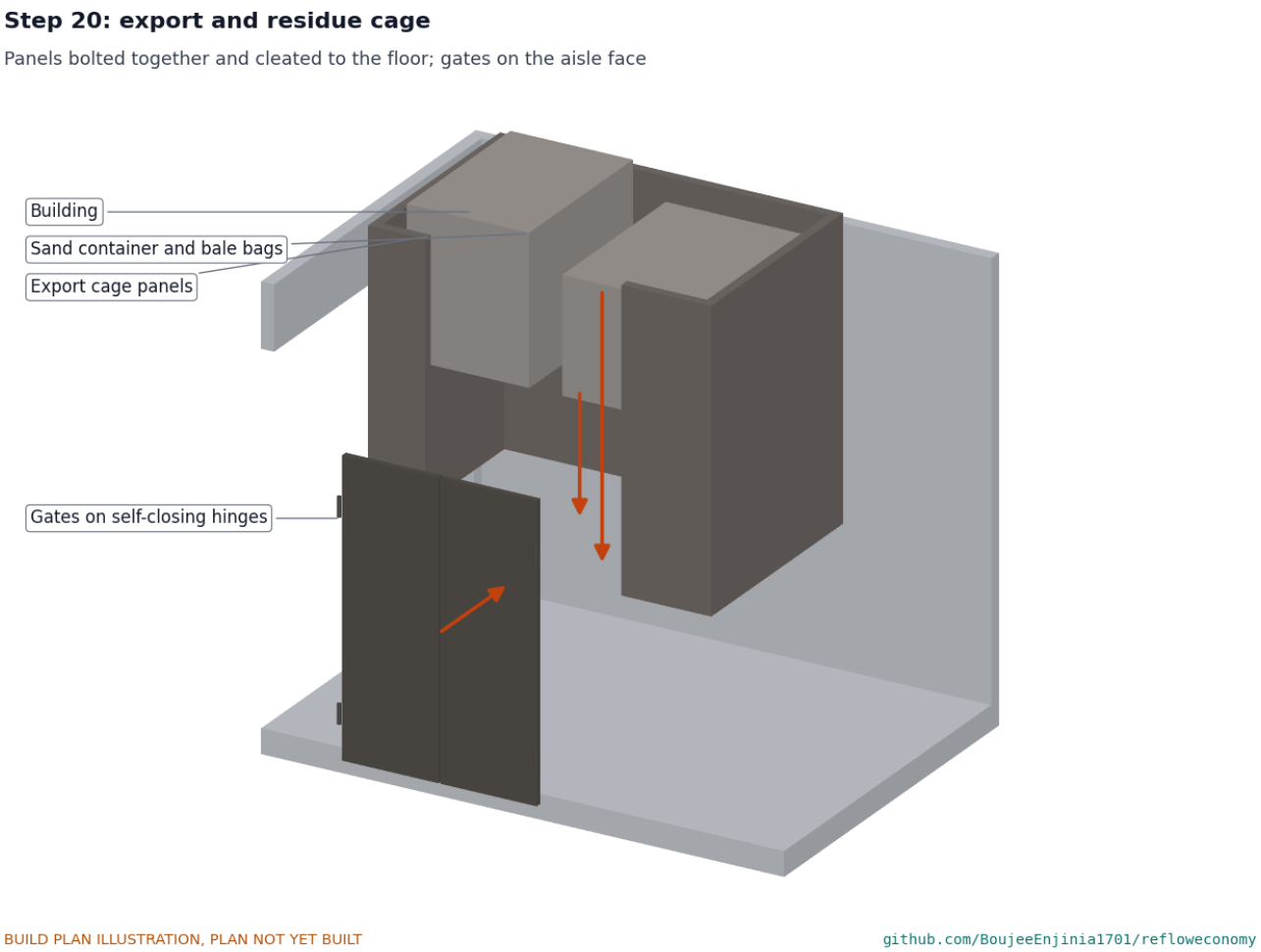


BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/refloweconomy

Racking anchored to the floor, bay 1 exactly 1.0 m from the hot zone line; desk at the product end; the third extinguisher beside the aisle.

Step 20: export and residue cage



Panels bolted together and cleated down; gates on self-closing hinges on the aisle face; sand container and bale bags inside.

5. First checks

These are the checks a TRL 4 test report would record; this plan only lists them. Requirement numbers are the product requirements of RFE-PRS-001 (PR-005 was R5 in the old requirements, and so on).

Table 2. First checks.

Check	Requirement	How	Pass when
Floor area and aisle	PR-005	Tape measure along the aisle and between rows	Clear aisle 1.0 m or more everywhere (1.57 m by design); nothing stands on the painted lines
Headroom over the aisle	PR-005	Measure under the tray crossing and the seam beam	2.0 m or more (2.02 m under the padding by design)
Hot zone clearance	PR-005	Measure from the boundary line to the nearest stock	1.0 m or more
Residual current devices	PR-012	Trip test on every circuit, by the electrician	Each trips within its rated time at 30 mA
Heater interlock	PR-012	Run the shredder, then try to switch on the press heaters; repeat with the extruder	The press heaters cannot switch on while either motor runs
Shredder interlocks and stop	PR-012	Open the feed slot flap, then the door, then press the emergency stop, each with the rotor running empty	The rotor stops each time and does not restart until reset
Hood face velocity	PR-011	Vane anemometer across opening 1 (sash over opening 2), opening 2 (sash over opening 1) and the hood B face, fan running, filters new	0.5 m/s or more averaged over each opening
Sash	PR-011	Slide it end to end	It covers each opening with 25 mm or more to spare and cannot leave both open
Shredder noise	PR-012	Sound level meter 1 m from the enclosure, shredder running on HDPE	Recorded; the area stays a hearing protection zone until this is known
Energy per shift	PR-009	Read the board's energy meter over a full reference shift	Recorded against 45.6 kWh
Supply current	PR-017	Clamp meter on the incoming supply through block A with every load running	At or under the rating of the supply breaker
Passport at the desk	PR-013, PR-015	Issue a passport for a test lot and time it	A valid record and label in 2 min or less

6. Safety stops

Stop at each point. Carry on only when everything listed is true.

- **S1. Before any container wall is cut.** The structural engineer's design and sequence are on site; the roofs are propped as it says; the seam beam is in and fixed. Nobody stands under a wall being cut.
- **S2. Before the supply is connected.** The electrician has installed and tested the board, the residual current devices, the emergency stop circuit and the heater interlock, and issued the installation certificate. Every socket near the washing tanks is residual-current protected and above splash height.
- **S3. Before any motor runs.** Guards on, emergency stops within reach of each machine, lockout padlocks and tags for every isolator, hearing protection at the shredder.

- **S4. Before any heater is switched on.** The fan runs; the face velocity at each opening has been measured at 0.5 m/s or more; the heater interlock has been tested; the two hot zone extinguishers are in place; nothing combustible is inside the hot zone line. Extraction runs whenever a heater is on and for 30 minutes after.
- **S5. Before the shredder takes material.** The flap and door interlocks stop the rotor every time; the door and the cage gates shut themselves; the intake rules are in force: no lithium cells, sharps, chemical containers or PVC in the feed; cells found at intake go straight into the sand container.
- **S6. Before the first melt.** Only identified HDPE and PP go into the extruder or press; PVC, polystyrene, PET and unknown plastics are never heated. Heat-resistant gloves and face shields are worn at the presses.
- **S7. Before anyone works a shift in it.** A qualified safety professional has reviewed the finished layout, the exits and the headroom against the local code, including the 2.02 m under the padded and taped tray crossing, the escape width past the open enclosure door (1.66 m by design) and the reach from the feed slot to the cutters measured on the built machine; workers are trained and none is under 18 (headroom and door rules decided by Amish, 2026-10-02).

7. Tools, skills and workspace

Tools. Angle grinder with cutting and flap discs; metal-cutting chop saw or bandsaw; MIG or stick welder with screens; bench drill and a magnetic drill or heavy hand drill; drills 3 to 13 mm and a countersink; rivet nut tool (M8 and M10); hand rivet tool; sheet metal shears and a folder for 1.5 mm sheet; jigsaw with a metal blade; hammer drill and anchor setting tool; spanners and sockets to M10; tape measure, chalk line, square and spirit level; floor paint rollers; vane anemometer; sound level meter; stopwatch. The electrician brings their own test instruments.

Skills. Structural steel welding to the engineer's drawings (a certified welder for the seam beam); general metalwork (marking out, cutting, drilling, welding light angle and tube, riveting and folding sheet); a licensed electrician for everything on the supply side; safe lifting with a crew or a pallet truck for the machines.

Workspace. The shell itself, with the doors open for ventilation while welding and painting; a welding area with screens and a fire watch for 30 minutes after hot work; level hard standing outside the back wall for the fan stand.

Personal protective equipment. Safety glasses and boots throughout; welding helmet, gloves and jacket; cut-resistant gloves for sheet and mesh; hearing protection for cutting and grinding; a hard hat while the roof is propped.

8. Where the numbers come from

- Model and constructability checks: `cad/src/model.py` (`python cad/src/model.py --check`); STEP and STL exports in `cad/step/` and `cad/stl/` .
- Pictures: `cad/src/build_plan_media.py` , using `.kit/build_views.py` ; written to `docs/05-build-plan/` and `cad/drawings/RFE-DWG-101` to `RFE-DWG-111` .
- General arrangement: `cad/drawings/RFE-DWG-001.pdf` , Rev P3.
- Calculations: `docs/04-calcs/01-sizing.md` (RFE-CAL-001 v0.3) and `docs/04-calcs/sizing.py` : layout and aisle (section 8), hoods and fan (section 6), noise (section 7), equipment cost (section 9).
- Bill of materials: `bom/bom.csv` and `bom/bom-notes.md` .
- Decisions: `docs/decisions/0003-design-for-construction.md` (RFE-DDR-003), with RFE-DDR-001 and RFE-DDR-002; open decisions in `docs/06-design-decisions.md` (RFE-DEC-001).
- Requirements: `docs/03-requirements.md` (RFE-REQ-001 v0.5).